

Katanin, kinesin-13 and ataxin-2 inhibit premature interaction between maternal and paternal genomes in *C. elegans* zygotes

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Abstract

Fertilization occurs before completion of oocyte meiosis in the majority of animal species and sperm contents move long distances within zygotes of mouse and *C. elegans*. If incorporated into the meiotic spindle, paternal chromosomes could be expelled into a polar body resulting in lethal monosomy. Through live imaging of fertilization in *C. elegans*, we found that the microtubule disassembling enzymes, katanin and kinesin-13 limit long range movement of sperm contents and that maternal ataxin-2 maintains paternal DNA and paternal mitochondria as a cohesive unit that moves together. Depletion of katanin or double depletion of kinesin-13 and ataxin-2 resulted in capture of the sperm contents by the meiotic spindle. Thus limiting movement of sperm contents and maintaining cohesion of sperm contents within the zygote both contribute to preventing premature interaction between maternal and paternal genomes.

eLife assessment

This is a **valuable** paper that identifies a potential challenge for embryos during fertilization: holding sperm contents in the fertilized embryos away from the oocyte meiotic spindle so that they don't get ejected into the polar body during meiotic chromosome segregation. The authors identify proteins involved in cytoplasmic streaming and maintaining the grouping of paternal organelles as being critical for this process. There remain minor weaknesses in the data presented but the paper provides **solid** evidence for the majority of its claims, and while the findings may pertain to a narrow audience the tools used and basic characterization shown will likely be relied upon by many in the community and therefore is of high value.

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Introduction

Reproduction by most animal species requires expulsion of chromosomes into polar bodies during oocyte meiosis to reduce chromosome number and fertilization by sperm to restore a diploid chromosome number. Because fertilization occurs during oocyte meiosis in most animal species, there is an inherent risk of sperm DNA being incorporated into the meiotic spindle and being expelled into a polar body. In some species, the first line of defense against this hypothetical calamity is ensuring that sperm does not fuse with the oocyte plasma membrane directly over the meiotic spindle. A Ran-GTP gradient emanating from the mouse metaphase II meiotic spindle excludes the fusion proteins Juno and CD9 from the oocyte plasma membrane over the spindle. When this mechanism was bypassed by injecting demembranated sperm adjacent to the spindle, the sperm DNA was ejected into a polar body (1). In *C. elegans*, the nucleus is positioned away from the site of future fertilization so that the meiosis I spindle assembles at the opposite end of the ellipsoid zygote from the site of fertilization (2–4).

However, simply controlling the site of fertilization is not sufficient to maintain a distance between the meiotic spindle and the sperm DNA because cytoplasmic streaming can move the sperm DNA long distances in both mouse (1) and *C. elegans* (5, 6). Meiotic cytoplasmic streaming in *C. elegans* embryos requires microtubules (7), kinesin-1 (3), and reticulons (8), but both mechanism and purpose are not completely understood. In addition to meiotic spindle microtubules, *C. elegans* meiotic embryos have cytoplasmic microtubules around the cortex and throughout the cytoplasm of the entire embryo (3). These microtubules are thought to drive meiotic cytoplasmic streaming because depletion of tubulin stops cytoplasmic streaming (7) and depletion of the microtubule-severing protein katanin by RNAi results in an increased mass of cortical microtubules and an increase in cytoplasmic streaming (8).

In addition to paternal DNA, *C. elegans* sperm introduce centrioles, SPE-11 protein, membranous organelles (MOs) and paternal mitochondria into the zygote. Paternal centrioles are silenced during meiosis by maternal KCA-1 (9), which is also required to pack yolk granules inward from the cortex (3) and the meiotic spindle outward toward the cortex (10). SPE-11 is an RNA-binding protein in sperm (11) that is required paternally for polar body extrusion (12) and proper embryonic development (13, 14). MOs are membrane vesicles in sperm that fuse with the sperm plasma membrane before fertilization during sperm activation (15). However, a subset of MOs that do not fuse with the sperm plasma membrane are introduced to the zygote at fertilization and are ubiquitinated with maternal ubiquitin (16). Paternal mitochondria are labelled with maternal autophagy machinery (17). Whereas both MOs and paternal mitochondria are eventually destroyed during embryogenesis, during meiosis they remain in a tight cloud around the sperm DNA (5, 16, 17). The mechanisms holding the sperm contents together in the zygote during cytoplasmic streaming have not been explored.

In this study we monitored the cloud of paternal mitochondria after increasing cytoplasmic streaming or disrupting the integrity of the cloud of paternal organelles. Our results suggest that both limiting cytoplasmic streaming and maintaining the integrity of the paternal organelle cloud contribute to preventing capture of the sperm contents by the meiotic spindle.

Results

Sperm derived DNA and mitochondria are maintained in a volume that excludes maternal mitochondria and yolk

granules but allows penetration by maternal ER during meiosis

Time-lapse in utero imaging of the meiotic spindle and an endoplasmic reticulum marker (ER) revealed that the ER is distributed throughout the zygote but appears as undulating lines during metaphase I (n=8), changes to a dispersed appearance during anaphase I (n=9), transitions back to undulating lines interspersed with large blobs during metaphase II (n=6), then changes to a dispersed pattern during anaphase II (n=5)([Fig. 1A](#) [↗](#); Video S1). Previous electron microscopy studies have indicated that the undulating lines correspond to sheet-like ER and the dispersed appearance corresponds to tubular ER ([18](#) [↗](#), [19](#) [↗](#)). A similar transition to sheet-like ER during mitotic M phase has been reported in HeLa cells ([20](#) [↗](#)) and *Xenopus* egg extracts ([21](#) [↗](#)), however, this may not be universal in all cell types ([22](#) [↗](#)). Pertinent to this study, ER morphology was used to determine cell-cycle stages during live imaging reported below in [Fig. 6](#) [↗](#).)

In striking contrast with the ER filling the entire zygote, yolk granules are packed inward, away from the cortex (n = 9 metaphase I; [Fig. 1B](#) [↗](#)) as previously described ([3](#) [↗](#)). When Mitotracker-treated males were mated to hermaphrodites expressing GFP-labeled yolk granules, the paternal mitochondria were found in a discrete cloud around the paternal DNA (n = 34; [Fig. 1B-C](#) [↗](#)) as previously described ([17](#) [↗](#)). Both the cloud of paternal mitochondria (n=19) and the meiotic spindle (n= 9 metaphase I) were observed in cortical regions that are free of yolk granules ([Fig. 1B](#) [↗](#)). Maternal mitochondria labeled with COX-4::GFP were also packed inward, away from the cortex, and were excluded from the cloud of paternal mitochondria and excluded from the spindle (n = 5 metaphase I; [Fig. 1C](#) [↗](#)). Time-lapse imaging (n=21) revealed that the cloud of paternal mitochondria remained together and separate from maternal yolk granules even when moving long distances with cytoplasmic streaming ([Fig. 1D](#) [↗](#)).

Because yolk granules and maternal mitochondria are packed inward during meiosis ([Fig. 1B, C](#) [↗](#)) and sperm must enter from the outside, it is possible that the exclusion of maternal yolk granules from the volume of paternal mitochondria might simply be a consequence of inward packing. In *kca-1(RNAi)* meiotic embryos, yolk granules ([3](#) [↗](#)) and mitochondria ([Fig. 2A, B](#) [↗](#)) do not pack and instead extend to the plasma membrane. Maternal mitochondria were still excluded from the volume of paternal mitochondria in *kca-1(RNAi)* embryos (n=11 control, 8 RNAi; [Fig. 2B](#) [↗](#)). This result indicated that the volume of paternal mitochondria excludes maternal mitochondria. In addition, paternal mitochondria also formed a discrete cluster in meiotic embryos of a nematode species with giant sperm ([23](#) [↗](#)) (n=4; [Fig. 2C](#) [↗](#)) indicating that the unique properties of the ball of paternal organelles in the meiotic zygote is conserved.

Maternal ER invades the volume of paternal organelles shortly after fertilization

In contrast with maternal yolk granules and maternal mitochondria, maternal ER was observed penetrating into the cloud of paternal mitochondria and enveloping the paternal DNA in a shell that appeared as a ring in confocal sections of both live and fixed meiotic embryos (n = 10; [Fig. 3A](#) [↗](#)). To elucidate how the maternal ER enters the ball of paternal organelles, we monitored 7 instances of sperm egg fusion by time-lapse microscopy. Previous studies have demonstrated that *C. elegans* sperm fuse with the egg as opposed to being phagocytosed ([24](#) [↗](#)). Mitotracker-labeled males were mated with hermaphrodites with a maternally-expressed ER marker and a maternally-expressed plasma membrane marker (mCherry::PH). 20 seconds after the apparent entry of paternal mitochondria into the egg (0:20 in [Fig. 3B](#) [↗](#)), a sperm-sized volume devoid of maternal ER was observed within the maternal plasma membrane. One minute after apparent sperm-egg fusion, the maternal ER had moved into the plasma membrane behind the sperm (1:25 in [Fig. 3B](#) [↗](#); 1:00 in [Fig. 3C](#) [↗](#)). The maternal ER ring enveloping the sperm DNA was not discernible until 7 min after apparent sperm-egg fusion (7:24 in [Fig. 3C](#) [↗](#)), although it might form earlier because the zygote undergoes dramatic movement through two sequential sphincters into the spermatheca then into the uterus during this time. The envelopment of the sperm DNA by

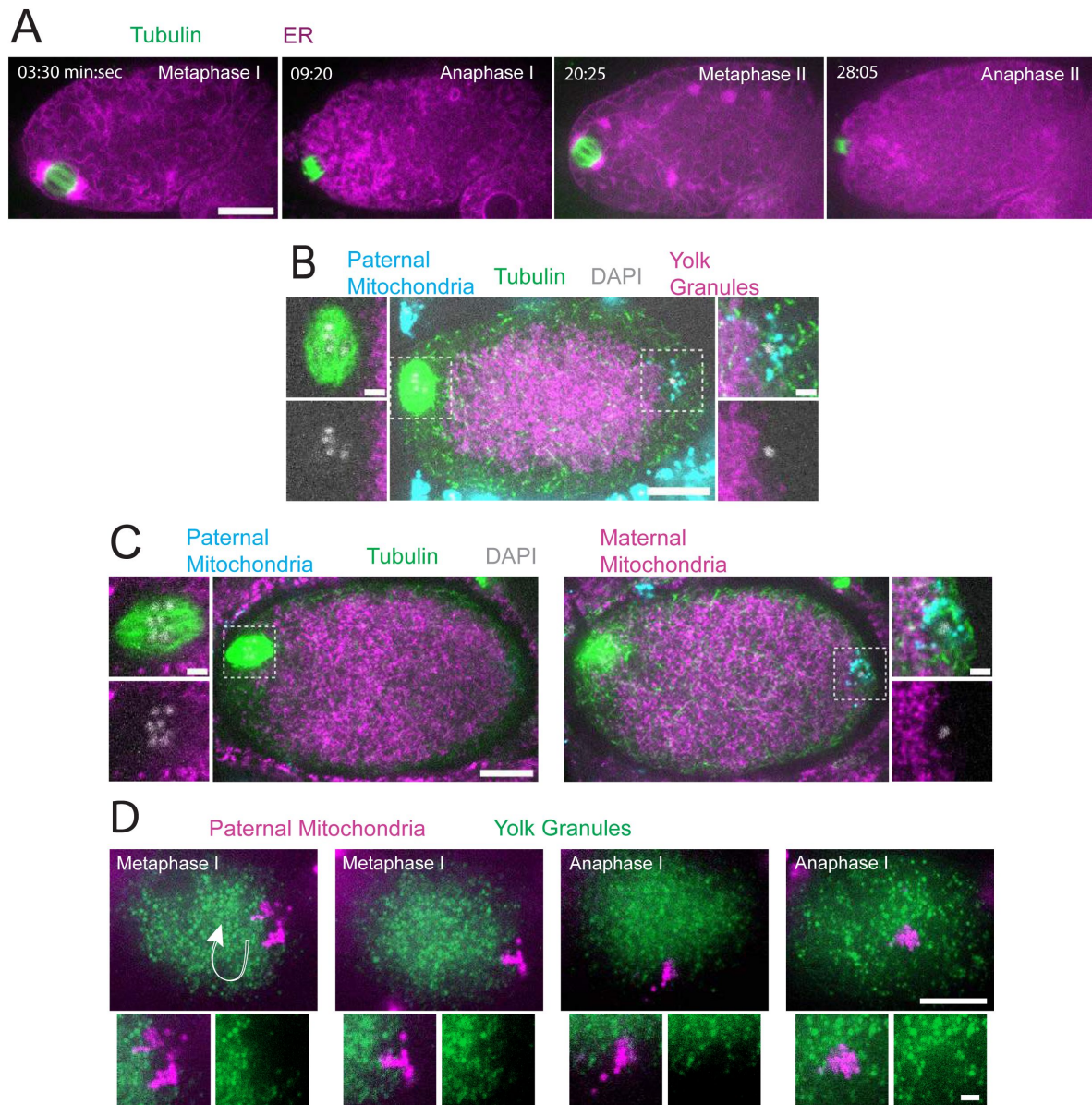


Figure 1.

Sperm contents exclude maternal yolk granules and mitochondria ER.

(A) Time-lapse in utero images of a control embryo expressing GFP::SPCS-1 (signal peptidase/ER) and mKate::TBA-2 (tubulin). ER morphology transitions from sheet-like during metaphase I and metaphase II to dispersed during anaphase I and anaphase II.

(B) Image of a fixed metaphase I embryo expressing VIT-2::GFP (maternal yolk granules), paternal mitochondria labelled with MitoTracker Deep Red FM, and stained with alpha tubulin antibody and DAPI. N=9 metaphase I embryos with packed yolk. (C) Image of a fixed metaphase I embryo expressing COX-4::GFP (maternal mitochondria), paternal mitochondria labeled with MitoTracker Deep Red FM, and stained with alpha tubulin antibody and DAPI. N=2 embryos with condensed maternal mitochondria. (D) Time-lapse in utero images of an embryo expressing VIT-2::GFP and paternal mitochondria labelled with SDHC-1::mCherry (succinate dehydrogenase). Images demonstrate sperm contents streaming in the short axis of the embryo. Arrow indicates direction of streaming. (A-D) Bars: whole embryos 10 μ m; insets 2 μ m.

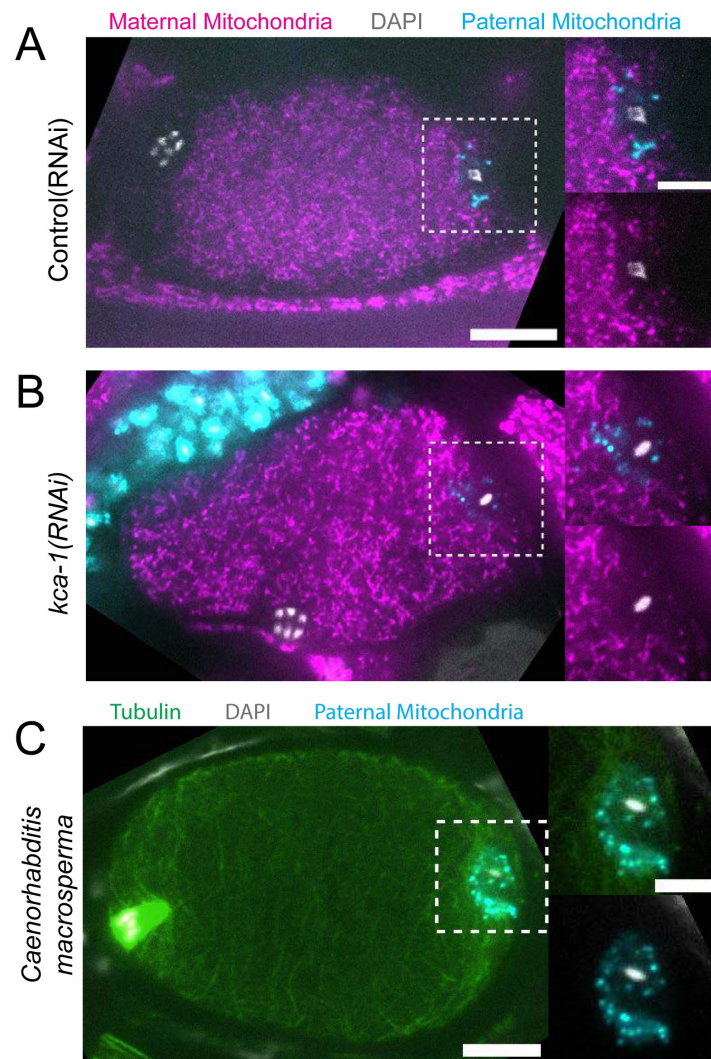


Figure 2.

Paternal organelle cloud is impermeable to maternal mitochondria regardless of yolk packing and is conserved in nematoda.

(A) Fixed image of a meiotic embryo expressing COX-4::GFP (maternal mitochondria) and MitoTracker Deep Red FM (paternal mitochondria). Paternal mitochondria take up a volume excluding maternal mitochondria (n=11). (B) Upon depletion of the kinesin cargo adapter, KCA-1, maternal mitochondria extend to the plasma membrane but paternal mitochondria remain sequestered (n=8). (C) Fixed image of a *Caenorhabditis macrosperma* (*C. macrosperma* wild isolate) meiotic embryo mated with MitoTracker Deep Red FM stained males. The paternal mitochondria are in a larger volume than in *C. elegans* but the cohesion of mitochondria near paternal DNA is conserved between species (n=4). (A-C) Bars: whole embryo 10 μ m; inset 5 μ m. White dotted boxes denote area of insets.

maternal ER may be an initial step in nuclear envelope assembly (25 [↗](#), 26 [↗](#)), which has been halted until after completion of meiosis. These results show that maternal ER is initially excluded from the sperm at fusion. Since maternal mitochondria and yolk granules are excluded later, this suggests that all maternal membranes are initially excluded from the sperm at fusion. The plasma membrane marker diffuses in behind the sperm first, followed by plasma-membrane associated ER, followed by envelopment of the sperm DNA by maternal ER by 7 min after fusion. Maternal yolk granules and maternal mitochondria, however, are excluded from the sperm volume for much longer (Fig. 1 [↗](#)).

The maternal ER envelope around the sperm DNA is permeable to proteins

If the maternal ER envelope around sperm DNA was sealed and impermeable during meiosis, this could both prevent the sperm DNA from inducing ectopic spindle assembly and prevent the sperm DNA from interacting with meiotic spindle microtubules. To test whether the ER envelope around the sperm DNA is permeable to proteins, we analyze BAF-1, a chromatin-binding component of the inner nuclear envelope (27 [↗](#)). Maternally provided GFP::BAF-1 was not detected on maternal or paternal chromatin during metaphase I or metaphase II but labelled the surface of both maternal and paternal chromatin during anaphase I and anaphase II (Fig. 4 [↗](#) A, B, C).

Because BAF-1 is a chromatin-binding protein, this result indicates that BAF-1 can pass freely through holes in the ER envelope surrounding the sperm DNA between metaphase I and anaphase I.

Movement of sperm contents within the zygote is limited by katanin and kinesin-13 and this limitation prevents capture of the sperm DNA by the meiotic spindle

C. elegans meiotic cytoplasmic streaming (3 [↗](#), 5 [↗](#), 7 [↗](#), 8 [↗](#)) has the potential to bring the sperm contents into close proximity with the meiotic spindle, however, a previous study found that the male pronucleus very rarely forms at the same end of the embryo as the female pronucleus (6 [↗](#)). This indicates that limitations on cytoplasmic streaming might be involved in maintaining distance between the sperm contents and meiotic spindle. We therefore asked whether increasing cytoplasmic streaming would cause collisions between the meiotic spindle and sperm contents. Meiotic cytoplasmic streaming requires microtubules (7 [↗](#)) and kinesin-1 (3 [↗](#)). Depletion of katanin by *mei-1(RNAi)* increases the mass of cortical microtubules and the extent of yolk granule streaming (8 [↗](#)) and depletion of kinesin-13 by *klp-7(RNAi)* also increases the mass of cortical microtubules (28 [↗](#)). We found that *mei-1(RNAi)* or *klp-7(RNAi)* increased the maximum displacement of the sperm contents in both the long and short axes of the ellipsoid embryo relative to control L4440(RNAi) embryos (Fig. 5A, B, C, D [↗](#); Videos S2, S3, S4). The only demonstrably significant difference between the excessive cytoplasmic streaming in these two depletions was that excessive streaming in *klp-7(RNAi)* persists for a longer period of time, well into metaphase II (Fig. 5D [↗](#)).

Among 15 control L4440(RNAi) time-lapse sequences, the closest center to center distance between spindle and sperm contents was 18 μm . In contrast, among 51 time-lapse sequences of *mei-1(RNAi)* meiotic embryos, the sperm came within 5.5 μm center to center distance of the spindle in 12 cases and in 12/12 of these cases, the sperm stopped moving relative to the spindle indicating a capture event (Fig. 5B [↗](#); Video S3). Among a subset that could be tracked through pronuclear stage, a single male pronucleus formed adjacent to multiple small female pronuclei. Among 25 time-lapse sequences of *klp-7(RNAi)* embryos, the sperm DNA became stuck against the meiotic spindle (4.6 μm center to center distance) in only one case. In all other cases, the sperm streamed past a stationary spindle. However, the closest sperm to spindle distance was 7.7 μm in one case. These

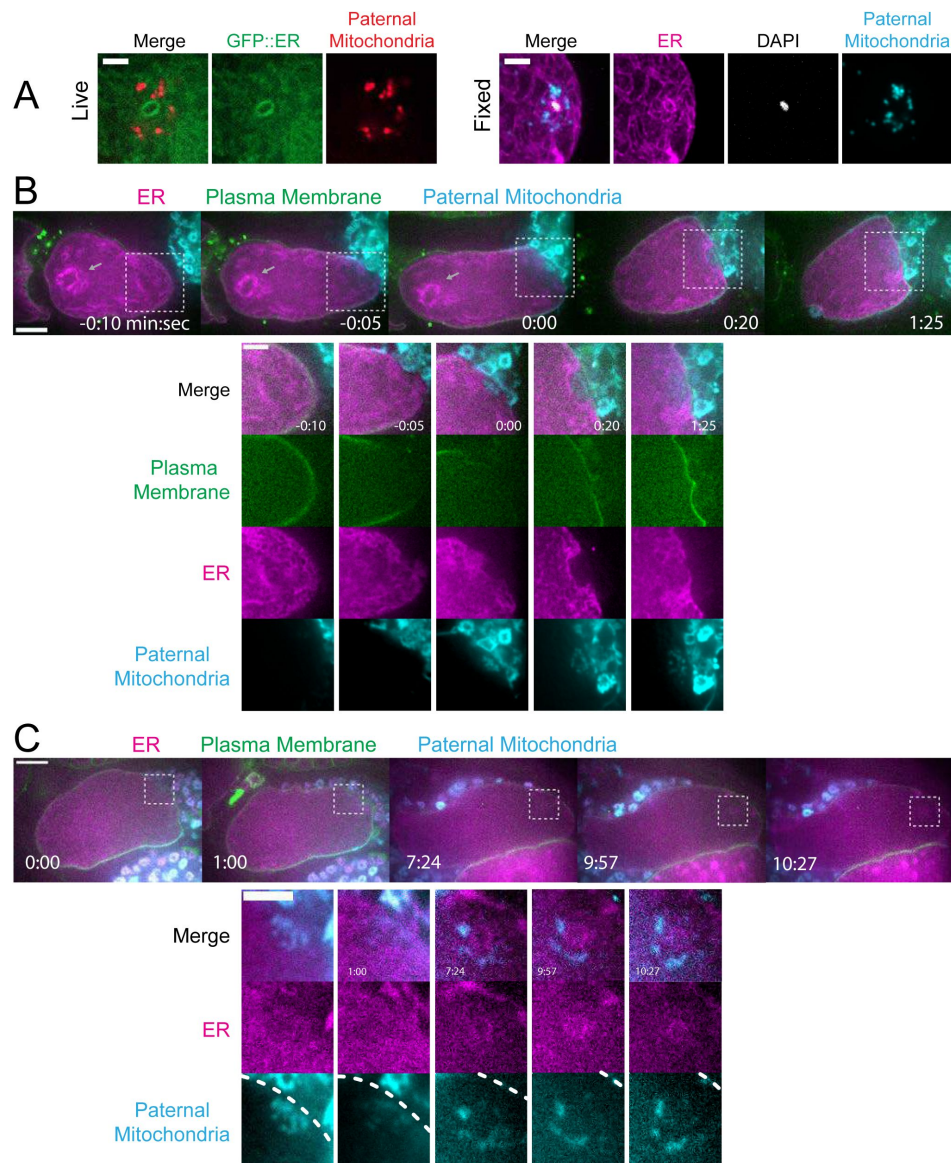


Figure 3.

Maternal ER enters the sperm cytoplasm after a delay and forms a ring around sperm DNA.

(A) Live and fixed images of anaphase I and metaphase I embryos show an ER ring within the sperm cytoplasm (as indicated by sperm mitochondria). The fixed image also shows DAPI staining within the ER ring. Live, $n=10$, fixed $n=10$. Bars: 3 μm (B) Time-lapse images of a strain expressing TMCO-1::GFP (maternal ER) and mCH::PH (maternal plasma membrane) and fertilized by a sperm labelled with MitoTracker Deep Red FM (paternal mitochondria). At fertilization, a gap appears in the MCH::PH as the sperm's membrane fuses with that of the oocyte. Paternal mitochondria are seen inside the oocyte at time 0:00. A "pocket" in the maternal ER contains the paternal mitochondria ($n=7$). At time 1:25 the mCH::PH gap has closed and fertilization is complete. Arrows denote prometaphase spindle. (C) Time-lapse images of an embryo as it enters the spermatheca and exits into the uterus. ER can be seen forming a ring within the mass of paternal mitochondria at 7:24 ($n=2$). Dotted lines denote cell membrane. White dotted boxes denote area of insets. (B-C) Bars: whole embryos 10 μm; insets 5 μm.

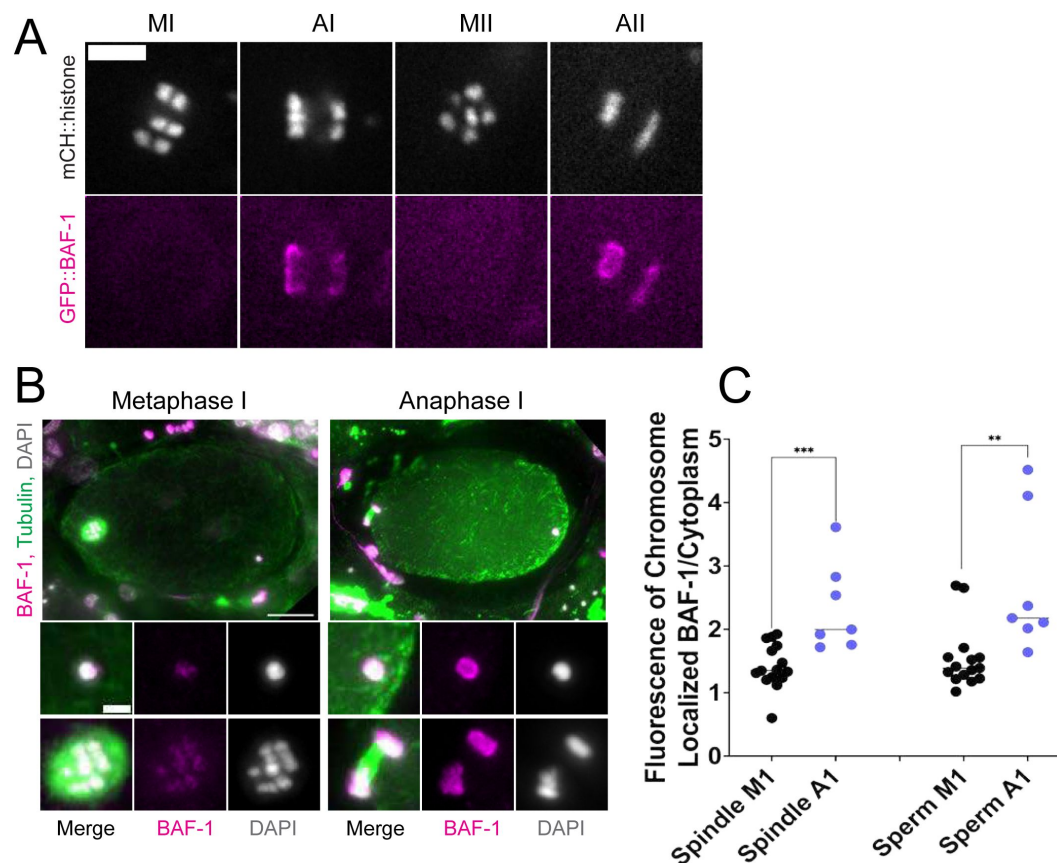


Figure 4.

ER surrounding the sperm DNA is permeable to the chromatin-binding protein BAF-1.

(A) Time-lapse images of maternal chromosomes in an embryo expressing mCH::HIS and GFP::BAF-1. Maternal BAF-1 localizes to chromosomes during anaphase, after the assembly of the ER envelope. Bar, 5 μ m. (B) In fixed embryos, maternal GFP::BAF-1 strongly localizes to both maternal and paternal chromosomes during anaphase I, but not metaphase I. Bars: whole embryo, 10 μ m; inset, 2 μ m. (C) Ratios of chromosomal to cytoplasmic GFP::BAF-1 show that, during anaphase I, there is an increase in GFP::BAF-1 on both the maternal and paternal chromosomes. ** $p < 0.01$, *** $p < 0.001$ by Mann-Whitney U Test.

results indicate that limiting cytoplasmic streaming is important for maintaining a distance between spindle and sperm and thus preventing capture of the sperm by the meiotic spindle. However, *mei-1(RNAi)* spindles are apolar and do not undergo normal polar body extrusion and the single capture event in a *klp-7(RNAi)* embryo resulted in cell cycle arrest. These results therefore do not reveal what would happen to paternal DNA captured by a normal meiotic spindle during polar body extrusion.

An important question is why the sperm contents move more than the meiotic spindle. During metaphase I, when the ER is sheet-like, movement of the sperm contents is limited. When the ER disperses during anaphase I, the spindle rotates and one pole is moved closer to the cortex by cytoplasmic dynein (Video S1; [\(29\)](#)). In contrast, the distance of the sperm contents from the cortex (Fig. S1) and movement of the sperm contents both increase when the ER disperses (Fig. 5C; Video S5).

Ataxin-2 is required to maintain cohesion of paternal mitochondria

Because maternal ER penetrates the sperm contents and envelops the sperm DNA, and because the *C. elegans* ortholog of ataxin-2, ATX-2, affects ER organization ([\(30\)](#)), we analyzed the contribution of ATX-2 to cohesion of the sperm contents in meiotic embryos. We first introduced an auxin-induced degron and GFP tag to the 3' end of the endogenous *atx-2* gene. Endogenously tagged ATX-2 was observed throughout oocytes and meiotic embryos (Fig. 6B, C; Fig. S2). ATX-2 did not uniquely co-localize with ER (Fig. S2). Dark holes were observed suggesting exclusion from the lumens of larger membranous organelles (Fig. 6C; Fig. S2). We then compared the intensity of the GFP signal after three different depletion treatments, 1 hour auxin, 24 hr *GFP(RNAi)* or 27 hr *atx-2(RNAi)*. All 3 methods resulted in significant reduction of the GFP signal (Figure 6B, 6C, 6D).

The extent of scattering of paternal mitochondria was analyzed in Z-stacks of fixed meiotic embryos depleted of ATX-2 by each of the 3 methods. This analysis was restricted to embryos from anaphase I through anaphase II because our streaming data (Fig. 5C) and that of Kimura ([\(6\)](#)) indicate that the sperm contents have not moved significantly before anaphase I. Example Z-projections are shown in Figure 7. Mitochondria can exist as a tubular network ([\(31\)](#)) and tubules in close proximity cannot always be resolved by light microscopy. To account for the apparent heterogeneous sizes of foci of paternal mitochondria, the fluorescence intensity of larger foci was divided by the fluorescence intensity of the smallest foci such that a focus with 3 times the fluorescence intensity was counted as 3 “mitochondria”. ATX-2 depletion by each of the 3 methods resulted in a significantly increased mean distance of paternal mitochondria from the sperm DNA (Figure 8A) as well as significantly increased standard deviation of individual distances (Figure 8B). Standard deviation is an important measure because an increase in the number of mitochondria both closer and further from the sperm DNA would result in no change in the mean. It remains possible that paternal mitochondria scatter is caused by pressure applied during fixation, pressure resulting from ovulation through the spermatheca valves, or cytoplasmic streaming. We were not able to unambiguously track scattering by live imaging because significant movement occurs during acquisition of a complete z stack and because the embryos cannot withstand the additional photodamage. These results still support the hypothesis that ATX-2 is required to maintain the integrity or cohesiveness of the ball of paternal mitochondria that surrounds the sperm DNA to resist external forces.

Double depletion of KLP-7 and ATX-2 results in capture of the sperm DNA by the spindle

Because we observed increased cytoplasmic streaming in *klp-7(RNAi)* and disruption of the integrity of the sperm contents after ATX-2 depletion, we hypothesized that double depletion of KLP-7 and ATX-2 would result in an increased frequency of spindle microtubules capturing the sperm DNA. Among 24 time-lapse sequences of *atx-2(AID+auxin) klp-7(RNAi)* meiotic embryos, the

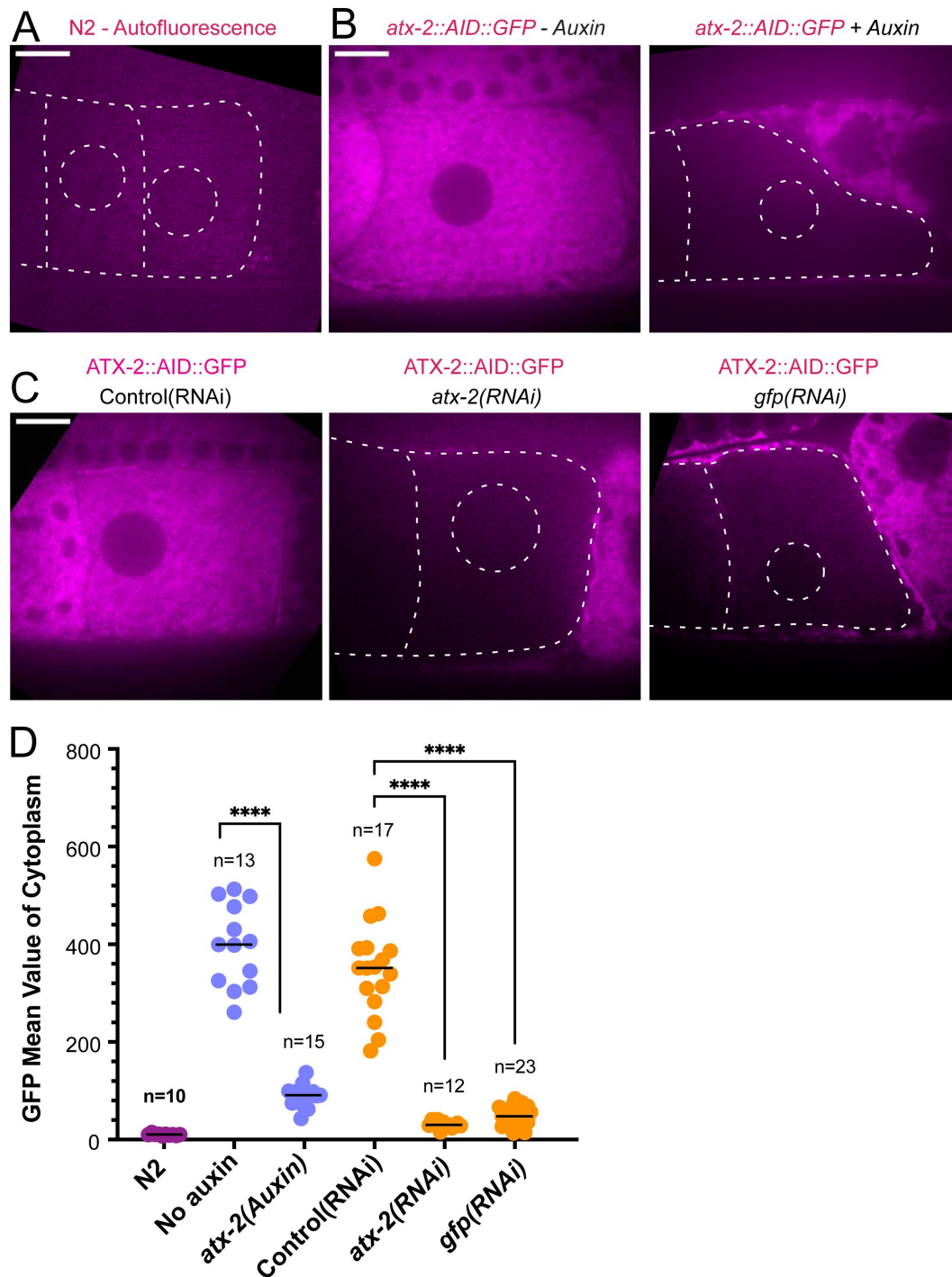


Figure 6.

ATX-2 is depleted by 3 different methods.

(A) Single plane live image of N2 (no GFP). Autofluorescence shows dim outline of oocytes and germinal vesicle. (B-C) Single plane live images of -1 oocytes in strain with *ATX-2::AID::GFP*. No auxin treatment shows endogenous GFP tagged ATX-2 fluorescence throughout the cytoplasm. Auxin treatment results in depletion of ATX-2 in oocytes. (C) Live images of -1 oocytes in strain with *ATX-2::AID::GFP*. Control L4440(RNAi) shows ATX-2 throughout the cytoplasm, but ATX-2 is depleted after 27 hr *atx-2*(RNAi) or 24 hr *gfp*(RNAi). (A-C) Bars, 10 μ m. (D) Mean GFP fluorescence in the cytoplasm of -1 oocytes after each treatment. **** $p < 0.0001$ by Welch's t-test and Brown-Forsythe test.

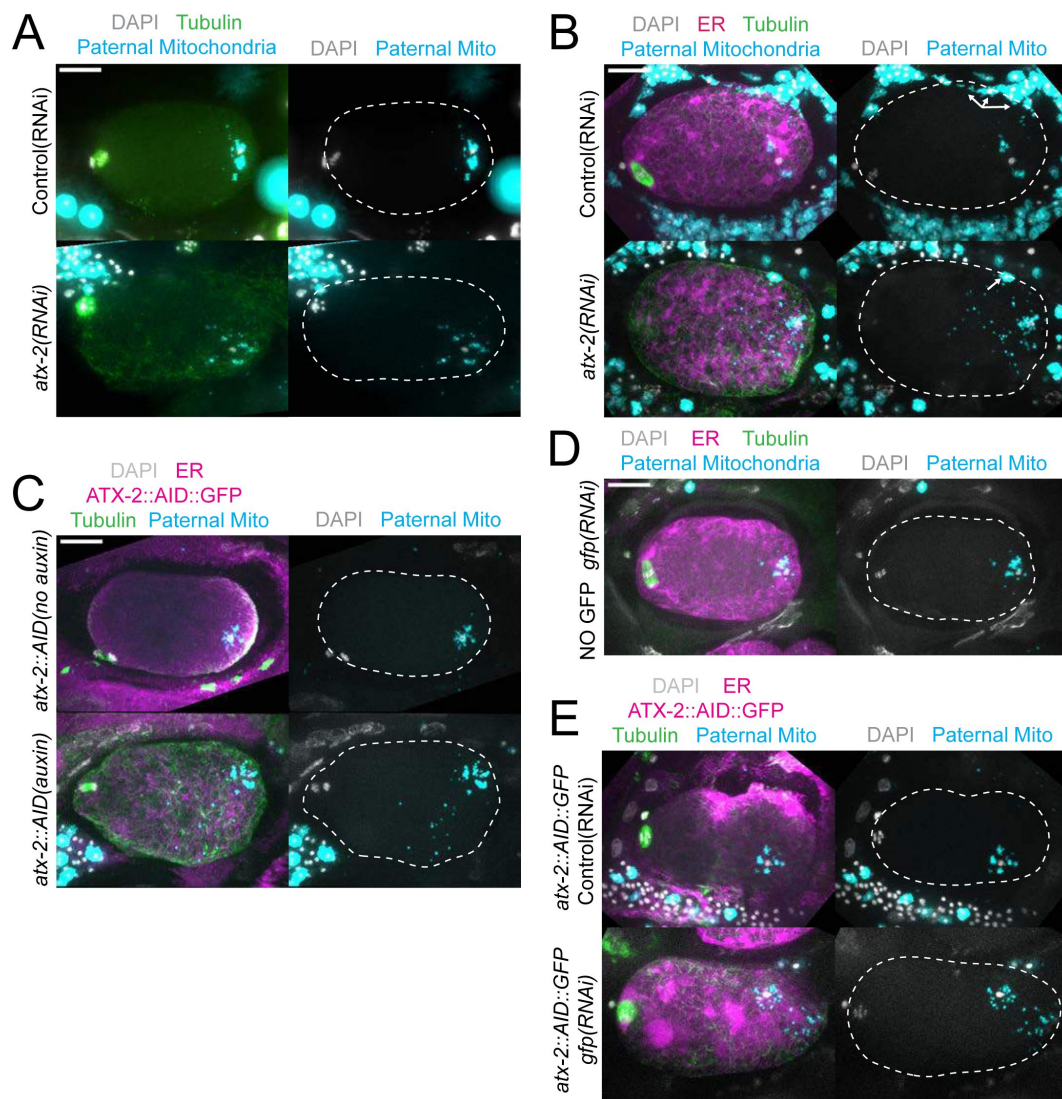
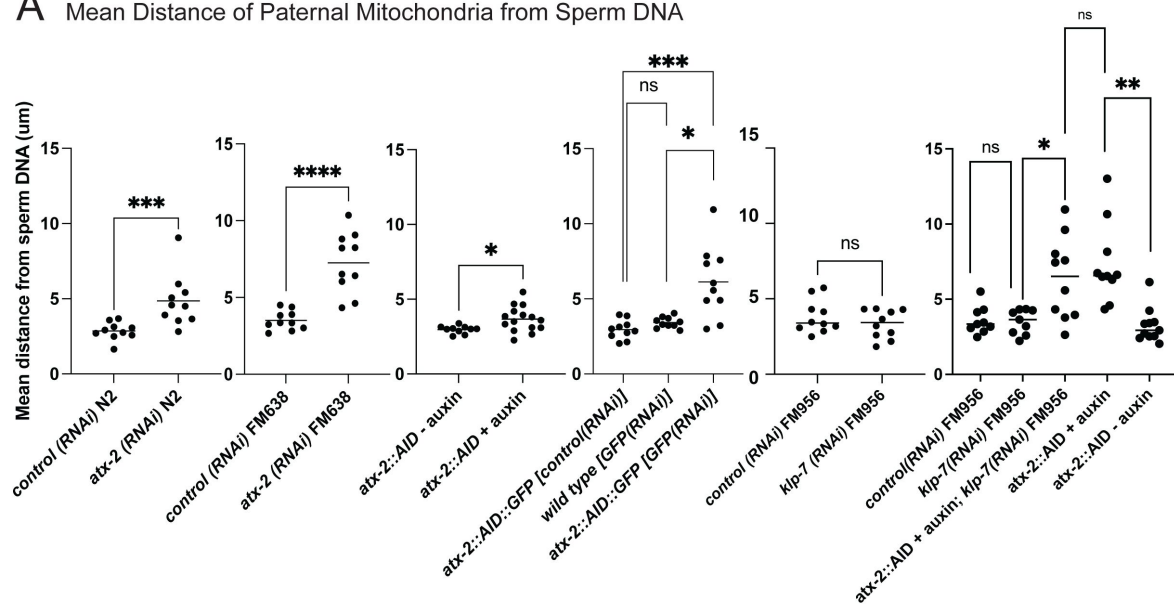


Figure 7.

Paternal mitochondria scatter during meiosis after ATX-2 depletion.

(A-E) Maximum intensity projections of z-stacks of fixed meiotic embryos stained with tubulin antibodies and DAPI and with paternal mitochondria labeled by mating with MitoTracker Deep Red FM treated *fog-2(q71)* males. (A) N2 wild-type embryos treated with control L4440(RNAi) or *atx-2(RNAi)*. (B, C, E) Embryos expressing TIR1, GFP::SPCS-1/ER, mKate::TBA-2, and with endogenously tagged ATX-2::AID::GFP. (B) 27 hr control L4440(RNAi) or *atx-2(RNAi)*. Arrows denote mitochondrial fluorescence from sperm outside the embryo overlapping with the embryo as a result of the maximum intensity projection. (C) No auxin or 1 hr auxin treatment. (D) GFP(RNAi) on strain with no tag on ATX-2 but expressing GFP::SPCS-1/ER and mKate::TBA-2. GFP::SPCS-1/ER fluorescence remains because SPCS-1 and ATX-2 are tagged with GFPs with different sequences. (E) Control L4440(RNAi) or *gfp(RNAi)* of ATX-2::AID::GFP strain. Bars, 10 μ m. White dotted outlines indicate the cortex of the cell.

A Mean Distance of Paternal Mitochondria from Sperm DNA



B Standard Deviation of Distance of Paternal Mitochondria from Sperm DNA

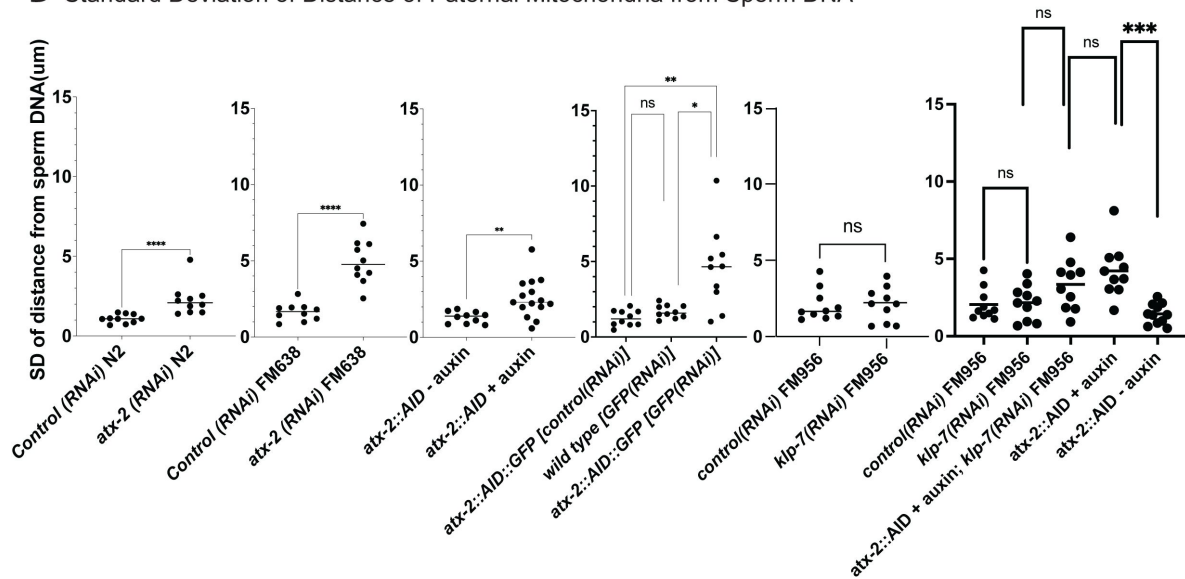


Figure 8.

Quantification of paternal mitochondrial scatter in ATX-2-depleted anaphase I meiotic embryos.

(A) Mean and (B) standard deviation of the distance of individual paternal mitochondria from the sperm DNA determined from Z-stacks of fixed anaphase I embryos. Each dot represents one embryo. Distances for individual mitochondria are in Supplementary data file.

ER envelope around the sperm DNA moved extensively in all cases but came within a threshold distance of 5.5 μm (center to center) of the meiotic spindle in only 5 cases. In 5/5 of these cases, a bundle of microtubules extended from the meiotic spindle into the ER envelope around the sperm DNA and the sperm DNA became stuck in this position (**Fig. 9A**, **9B**). In one case, the ER envelope around the sperm DNA was transiently captured by the meiosis I spindle and stretched in the direction of streaming (**Fig. 9C**). The sperm DNA then released, continued streaming and was captured by the meiosis II spindle. All cases of stable capture caused a cell-cycle arrest so that the consequences on polar body extrusion could not be determined.

Because ATX-2 depletion alters ER morphology, we were not able to score cytoplasmic streaming with the cell-cycle accuracy shown in **Fig. 5**. However, the maximum long axis and short axis displacement at any cell-cycle time was increased in *atx-2(AID)* with auxin vs without auxin (**Fig. S3**). No spindle capture events were observed, however, among 15 ATX-2 single depletion time-lapse sequences and the closest distance between sperm and spindle was 8.2 μm with 1 hr auxin and 19.3 μm without auxin.

The frequency of sperm capture by the meiotic spindle (**Fig. 9D**) was significantly higher than wild-type controls in *klp-7(RNAi) atx-2(AID)* double depleted embryos ($p=0.011$ Fisher's exact test). Although the number of single mutant embryos analyzed was too low to demonstrate a significant difference between single and double mutant embryos, these results qualitatively support the hypothesis that limiting cytoplasmic streaming and maintaining the integrity of the ball of paternal mitochondria are both important for preventing capture events between the meiotic spindle and sperm DNA.

Discussion

The mechanism excluding maternal yolk granules and mitochondria from the volume of sperm cytoplasm introduced to the egg at fertilization is not clear. The simplest hypothesis is that maternal and paternal cytoplasm might not mix during the 45 min from GVBD to pronucleus formation due to the high viscosity of cytoplasm. Attempts at measuring cytoplasmic viscosity of the *C. elegans* zygote have revealed values from 0.67 – 1.0 Pa s (32–34) which are similar to the viscosity of 100% glycerol. Alternatively, the sperm contents might be held together by a cytoskeleton-like matrix as proposed for the Balbiani body (35). In either case, an active process appears to allow the maternal ER to penetrate into the paternal cytoplasm to envelope the sperm DNA. The capture of the sperm DNA by the meiotic spindle in ATX-2 KLP-7 double depleted embryos (**Fig. 9**) suggests that the integrity of the exclusion zone around the sperm DNA might insulate the sperm DNA from spindle microtubules. However, a much larger number of *klp-7(RNAi)* singly depleted and *atx-2(degron)* singly depleted time-lapse sequences are needed to rigorously support this idea.

ATX-2 is required to maintain the integrity of the ball of paternal mitochondria around the sperm DNA (**Figs. 7** and **8**), but the mechanism is unknown. Because the paternal mitochondria observed to scatter are from wild-type males, the effect of ATX-2 depletion must be on the egg and not on the sperm. Although ATX-2 depletion alters ER morphology (30) we still observed a maternal ER envelope around the sperm DNA in all ATX-2-depleted embryos. ATX-2 also plays roles in translational regulation (36), germline proliferation (37), cytokinesis (38), centrosome size (39), and fat metabolism (40). Thus the effects of ATX-2 could be extremely indirect. Because *C. elegans* ovulate every 23 min (4), however, our rapid 1 hr depletion would only affect the three most mature oocytes. A speculative possibility is that ATX-2 regulates “the integrity of the cytoplasm” analogous to the action of ANC-1 in the *C. elegans* hypodermis (41). Because ATX-2 is an RNA-binding protein with intrinsically disordered domains, it might act as part of the RNA-dependent ER-associated TIS granule network (42, 43).

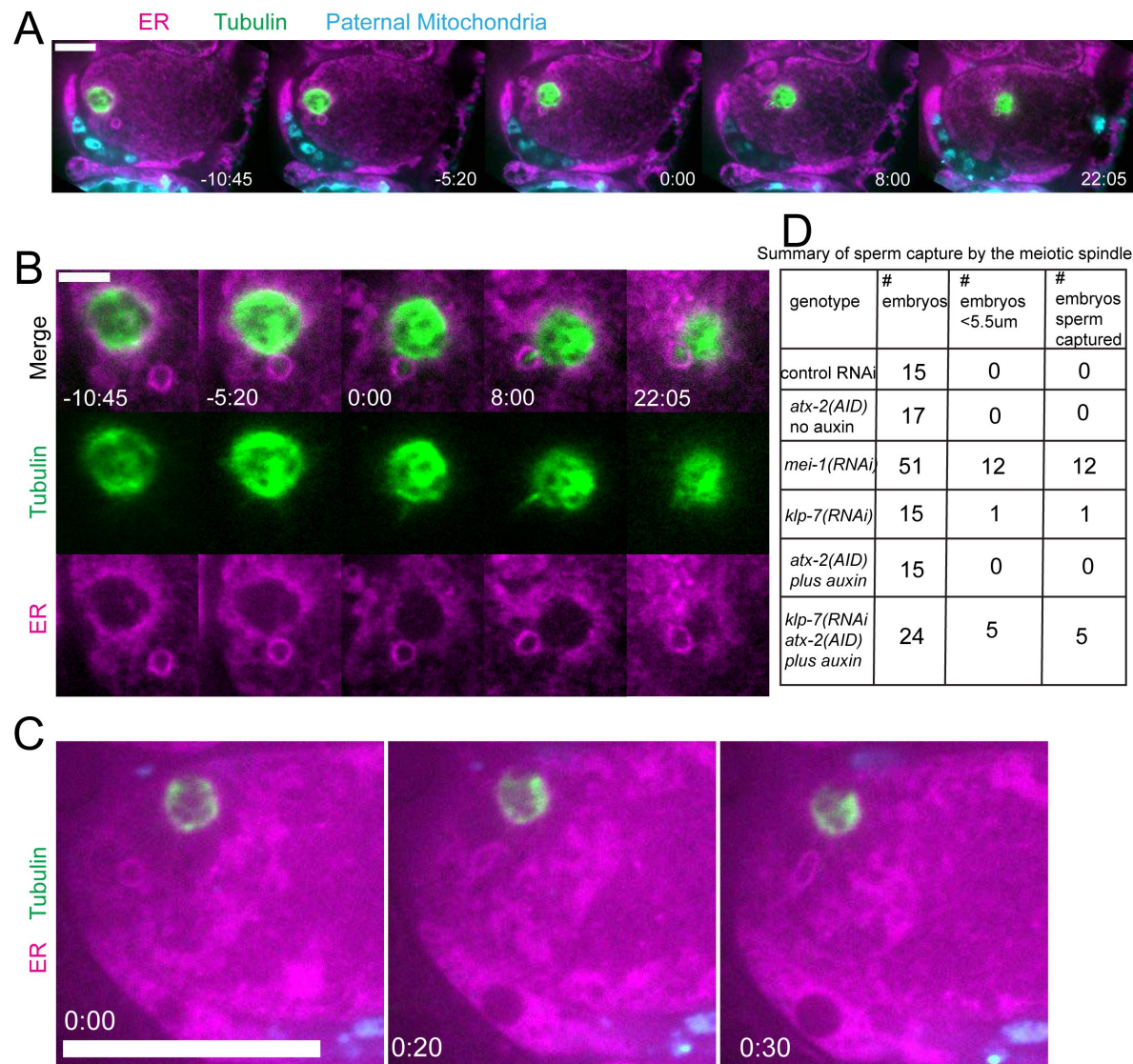


Figure 9.

Capture of the sperm DNA by the meiotic spindle in KLP-7 ATX-2 double depleted meiotic embryos.

(A) Live imaging of GFP::SP12/ER; mKATE::TUB; ATX-2::AID::GFP meiotic embryo with Deep Red MitoTracker stained paternal mitochondria from *fog-2(q71)* males mated in. Hermaphrodites were treated with *atx-2/klp-7 (auxin/RNAi)*. 5/24 videos had sperm ring travel within <5.5 um of spindle (measured center to center). Of those videos, 5/5 resulted in a microtubule bridge between the spindle and the middle of the sperm ring. (B) Higher magnification of (A). (C) Example of ER ring around the sperm DNA stretching toward the spindle. (D) Summary of capture events. Bars: (whole embryo) 10 um; (inset) 5 um.

In control embryos, the sperm contents rarely came near the meiotic spindle (18 μm closest distance, see text above) in agreement with a previous study that found that male and female pronuclei rarely form next to each other (6). Streaming of the sperm contents was most commonly restricted to a jostling motion with little net displacement, circular streaming in the short axis of the embryo, or long axis streaming in which the sperm turned away from the spindle before the halfway point of the embryo (Fig. 5). Depletion of MEI-1 or KLP-7 resulted in longer excursions of the sperm contents in the long axis of the embryo toward the spindle (Fig. 5) but frequent capture of the sperm by the spindle was only observed in *mei-1(RNAi)* (Fig. 5 and 9D). This may be because *mei-1(RNAi)* affects positioning of the spindle within the embryo (7) or because the altered structure of the *mei-1(RNAi)* spindle allows spindle microtubules to capture chromosomes further from the spindle center. In capture events observed after double depletion of ATX-2 and KLP-7, a bundle of microtubules was discernible extending from the spindle into the ER envelope surrounding the sperm DNA (Fig. 9). Such bundles were not observed in *mei-1(RNAi)* capture events (Fig. 5), likely because of the previously reported low density of microtubules in *mei-1(RNAi)* spindles (44, 45).

To our knowledge, the only reported example of premature interaction between the sperm contents and the oocyte meiotic spindle outside of *C. elegans* is the extrusion of sperm DNA into the second polar body when sperm was injected adjacent to the mouse metaphase II spindle (1). Close proximity of the sperm contents to the meiotic spindle and apparent capture events in *C. elegans* have been reported in only 20-25% of *mei-1(RNAi)*, *klp-7 atx-2* double depletion (Fig. 9D) or *kca-1(RNAi)* (9) meiotic embryos. In the case of *kca-1(RNAi)*, premature sperm asters were implicated in the capture events (9), however, *kca-1(RNAi)* blocks cytoplasmic streaming (3) and thus likely reduces the probability of the sperm contents moving close to the spindle, which is also mispositioned in *kca-1(RNAi)* (10). Close proximity of the sperm DNA with the meiotic spindle was also reported in a small fraction of fixed *pfn-1(RNAi)* embryos where increased mobility of the sperm contents was also reported but actual capture events were not reported (5). It should be noted that the large volume of oocytes may contribute to the rarity of sperm/spindle capture events. An 80 μm diameter mouse zygote has a volume 16 times greater than a 50 μm x 25 μm ellipsoid *C. elegans* zygote and a 120 μm diameter human zygote has a volume 55 times greater than a *C. elegans* zygote [using the equation $\frac{4}{3} \pi (r^a)(r^b)(r^c)$].

None of our time-lapse sequences of sperm capture by the meiotic spindle resulted in extrusion of paternal DNA into a polar body as was observed when sperm was injected next to the meiotic spindle of mouse oocytes (1), likely because of the pleiotropic effects of depleting MEI-1, KLP-7 or ATX-2. Many of the reported *kca-1(RNAi)* capture events also resulted in arrest (9). Future development of more specific perturbations of cytoplasmic streaming and the organelle exclusion zone around the sperm DNA should address this problem.

Materials and Methods

C. elegans Strains

Genotypes of strains used in this study are listed in Table S1.

Live *in utero* imaging

Adult hermaphrodites were anaesthetized with tricaine/tetramisole in PBS as described (4, 46) and then placed on 2% agarose pads on slides. Extra anesthetic was gently pipetted onto the agarose pad and a coverslip was placed on top. The slide was inverted and placed on the stage of an inverted microscope. Meiotic embryos were identified by bright-field microscopy before initiating time-lapse fluorescence. For all live imaging, the stage and immersion oil temperature were 21°C-24°C. For all time-lapse data, single-focal plane images were acquired with a Solamere spinning disk confocal microscope equipped with an Olympus IX-70 stand, Yokogawa CSU10,

either Hamamatsu ORCA FLASH 4.0 CMOS (complementary metal oxide semiconductor) detector or Hamamatsu ORCA-Quest qCMOS (quantitative complementary metal oxide semiconductor) detector, Olympus 100x UPlanApo1.35 oil objective, 100-mW Coherent Obis lasers (405, 640, 488, 561 nm) set at 30% power, and MicroManager software control. Pixel size was 65nm for the ORCA FLASH 4.0 CMOS detector and 46nm for the ORCA-Quest qCMOS detector. Exposures were 200ms for the ORCA FLASH 4.0 qCMOS detector and 100ms for the ORCA-Quest qCMOS detector. Time interval between image pairs or trios was 5 seconds. Focus was adjusted manually during time-lapse imaging.

For the ATX-2 images in **Fig. 6**, z-stacks of ~1 oocytes of anesthetized live worms were captured with a Zeiss LSM 980 confocal microscope with Airyscan 2 and a Zeiss Objective LD LCI Plan-Apochromat 40x/1.2 Imm Corr DIC M27 for water, silicon oil or glycerine.

Fixed Immunofluorescence

C. elegans meiotic embryos were extruded from hermaphrodites in 0.8× egg buffer by gently compressing worms between a coverslip and a slide, flash frozen in liquid N₂, permeabilized by removing the coverslip, and then fixed in ice-cold methanol before staining with antibodies and DAPI. The primary antibodies used in this work were mouse monoclonal anti-tubulin (DM1α; Thermo Fisher Scientific; 1:200) and rabbit anti-GFP (NB600-308; Novus Biologicals; 1:600). The secondary antibodies used were Alexa Fluor 488 anti-rabbit (A-21206; Thermo Fisher Scientific; 1:200), Alexa Fluor 488 anti-mouse (A-21202; Thermo Fisher Scientific; 1:200) and Alexa Fluor 594 anti-mouse (A-21203; Thermo Fisher Scientific; 1:200). Z-stacks were captured at 1-μm steps for meiotic embryos using the same microscope described above for live imaging.

Sperm ER Ring Filming

For sperm ER ring filming, the ring was observed throughout meiosis I and II and manually kept in the focal plane by adjusting the stage (z axis). Filming typically began at spermatheca exit or meiosis I and ended at pronuclear formation or cell arrest. Due to the rigors of filming an embryo with optimal orientation and positioning in the uterus, videos that started and ended mid phases were still filmed and included in measurements so long as a complete phase was included between starting and ending filming (e.g., AI and MII were measured in videos starting mid MI and ending mid AII). In these cases when filming started mid phase and/or ended prematurely mid phase, the incompletely filmed phases were not included in quantifications. The phases at which cell arrest occurred were not included in quantifications, but the phases prior were still used.

Sperm ER Ring Tracking

Measurements of the ER rings' dynamics in videos of embryos were tracked manually using Fiji plugin MTrackJ. All embryos quantified were rotated such that at the beginning of each video the pole containing the meiotic spindle was at the left. Only embryos measured at >35 μm long were used for ER ring tracking. Embryos measured <35 μm were tilted and not ideal for tracking. The starting and ending frames of each cell phase were then determined based on ER morphology. Individual tracks for each phase were manually made by clicking on the center of the ER ring, or in cases when it was briefly out of focus, the mitochondria that most closely followed where the ER ring was previously seen. The following criteria was used for determining phases to be quantified by tracks:

Metaphase I: Starts when embryo becomes stationary after exiting spermatheca into uterus and ends when ER reticulation becomes dispersed.

Anaphase I: Starts when ER becomes dispersed and ends when it begins to reticulate.

Metaphase II: Starts when ER begins to reticulate and ends when it becomes dispersed.

Anaphase II: Starts when ER becomes dispersed and ends at paternal pronuclear formation.

Maximum X-axis displacement was quantified by subtracting the minimum x-axis coordinate of a phase's track from the maximum x-axis coordinate.

Maximum Y-axis displacement was quantified by subtracting the minimum y-axis coordinate of a phase's track from the maximum y-axis coordinate.

Displacement was measured as the distance between the first point of a phase's track and the last point.

Distance Traveled was measured as the total length of the track that the sperm ring traveled.

Maximum Average Velocity was measured by taking the maximum of 3 point moving averages of velocities in each phase. MTrackJ was used to measure the velocity of the ER ring between each video frame.

Duration of Phases were calculated by subtracting the first frame number of a phase from the last one and multiplying by 5 due to the 5 second intervals.

Paternal Mitochondria Scattering Quantifications

Paternal mitochondrial scattering was calculated by using a circular ROI to measure the mean value fluorescence of Deep Red MitoTracker FM or Red MitoTracker CMXros labeling paternal mitochondria in meiotic embryos. Mean pixel values were taken using a circle ROI with an 18-pixel diameter to cover the entire area of a region of fluorescence that appeared as a punctum (unless specified otherwise due to a different resolution). This was repeated until all puncta fluorescence in an embryo were measured. For every embryo, a region of cytoplasm without mitochondria had its fluorescence measured. This value was subtracted from every mitochondrial measurement to correct for noise. The distances between the center of each punctum and the center of the paternal DNA were then recorded with a line tool. The Pythagorean theorem was used to determine the distance when the mitochondria were in a different z-stack than the paternal DNA. These distances were recorded in column scatter graphs to observe the distribution of mitochondrial distance from sperm DNA quantitatively. To measure large amorphous masses of mitochondria, the same sized ROI measuring fluorescence in distinct puncta was used to cover a portion of the mass, make measurements, and then moved to another portion to make more measurements. This was repeated until the entire area of the mass had its fluorescence measured. The corresponding distances to the paternal DNA were recorded for every ROI used. The mean value fluorescence of the amorphous masses were typically much greater than the distinct puncta and as such presumably had a greater density of mitochondria per ROI than the puncta. In order to account for this density, the mean values measured from the ROI's over the masses were divided by the average of the mean values of all of the puncta in an embryo. This resulting number was then used to determine how many times the distance between the paternal DNA and the ROI was recorded in the column scatter graphs (e.g., if the rounded value was 2 then the distance was recorded twice). The averages and standard deviations of each distribution of mitochondrial distances from paternal DNA in the embryos were then measured in order to compare the scattering of mitochondria in different experimental treatments. Quantifications were done in meiotic embryos of all phases except metaphase I. Since this phase is right after fertilization, we believe embryos do not have the time to exhibit scattering as a phenotype.

Paternal Mitochondria Labeling

L4 *fog-2(q71)* males were picked onto an OP50 plate and treated with 200 μ L 0.05 mM working stock of MitoTracker Deep Red FM (Invitrogen) or MitoTracker Red CMXros (Invitrogen) in M9 overnight. Subsequently all the males were moved to a fresh OP50 plate and incubated for 15

minutes to “wash” away excess MitoTracker. This washing step was conducted three times to fully remove excess MitoTracker. After the third wash, the males were then moved to plates with hermaphrodites 24 hours before they were to be filmed or used for immunofluorescence. A minimum ratio of 1:1 males to hermaphrodites was used for matings.

L4 *sdhc-1::mCherry; him-5(e1490)* males were added to plates with hermaphrodites 24 hours before they were to be filmed. A minimum ratio of 1:1 males to hermaphrodites was used for matings.

HALO Ligand

At least 20 hrs before imaging, hermaphrodites expressing HaloTag were treated with 2.5 μ M Janelia Fluor HaloTag Ligand 646 or 549 in M9 added to the bacterial lawn of an MYOB agar plate.

RNA interference

For RNA interference, L4 hermaphrodites were placed on RNAi plates with an RNAi bacterial lawn for a set period of time before being used for live imaging or fixed slides. In 48 hrs treatment, the worms were moved to a fresh RNAi plate after 24 hrs. RNAi plates were always seeded the day before adding worms. For *mei-1*, *atx-2*, *gfp*, and L4440 (*RNAi*), worms fed on RNAi bacterial lawns for 24-28 hrs. For *klp-7* (*RNAi*) and its corresponding control L4440 (*RNAi*), worms fed on RNAi bacterial lawns for 48-52 hrs.

Auxin induced degradation

For auxin induced degradation, L4's of strains endogenously tagged with auxin-inducible degrons and a TIR1 transgene were placed on a fresh plate of OP50. After 24 hrs, the hermaphrodites were moved to auxin plates with lawns of OP50 for 1-3 hrs before use in live or fixed experiments. 4 mM auxin plates were made by adding 400 mM auxin (indole acetic acid) in ethanol to molten agar which was then poured and seeded with OP50 bacteria. Depletion of ATX-2::AID::GFP was confirmed by the reduction of ATX-2::AID::GFP signal in –1 oocytes.

Double RNA interference and auxin induced degradation

In AID *klp-7* *RNAi* experiments, L4's were placed on *klp-7* (*RNAi*) and then transferred to a fresh *klp-7* (*RNAi*) lawn after 24 hrs. After a total of 47 hrs of treatment, the worms were moved to auxin/*RNAi* plates seeded with *klp-7* (*RNAi*). 1 hr later the worms were then used in live or fixed experiments. Auxin/*RNAi* plates consisted of 4 mM auxin, 1 mM IPTG, and 200 μ g/mL ampicillin in agar. Stocks of 400 mM auxin, 1 M IPTG, and 200 mg/mL ampicillin were added to molten agar and mixed to create the plates.

Fluorescence intensity measurements

The depletion of *atx-2* was measured by taking the mean value of an ROI over the cytoplasm of –1 oocytes, taking caution to not include the nucleus. The z-stack at which the nucleus appeared most in focus was used for measurements in each embryo. The mean value of the cytoplasm was then subtracted by the mean value of an area not containing any part of the worm to correct for noise.

GFP::BAF-1 fluorescence was measured by manually tracing the outline of chromosomes during metaphase I and II and then taking the mean value fluorescence. The same ROI was also used to measure the mean value fluorescence of the middle of the embryo's cytoplasm. The mean value of the fluorescence over the chromosomes was then divided by the cytoplasmic mean value in order to measure and compare BAF-1 fluorescence in metaphase I vs anaphase I

Statistics

Shapiro-Wilks tests through GraphPad Prism were used to test for normality in all data in which statistical tests were used to compare means. If the test determined the data was normal, P values were calculated in GraphPad Prism using Welch's T tests for comparing means of only two groups and ANOVA tests for comparing means of three or more groups. If the data was not normal, P values were calculated in GraphPad Prism using Mann-Whitney tests for comparing means of only two groups and Kruskal-Wallis tests for comparing means of three or more groups.

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Supplemental Data

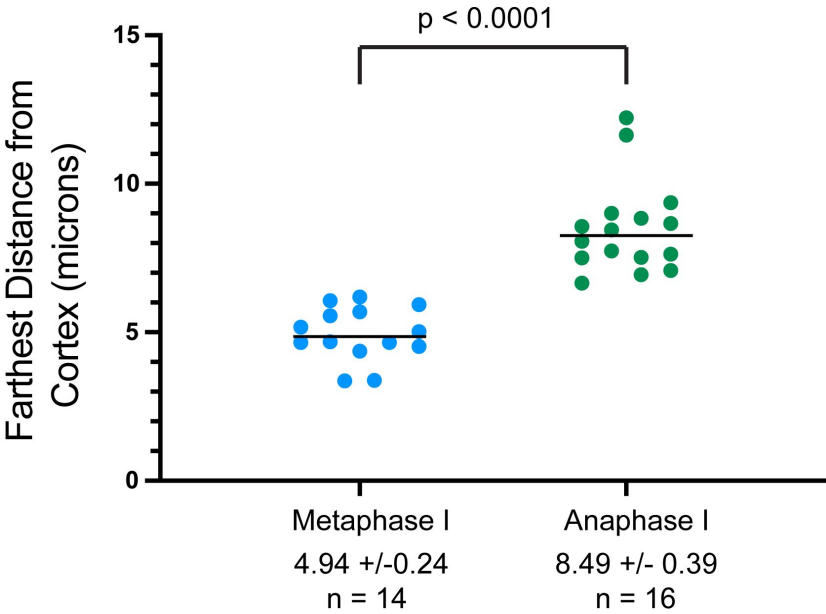
Table S1

C. elegans strains used in this study.

Strain Name	Other Name	Genotype Description
N2 Bristol		
FM111	WH327	<i>unc-119(ed3) III; ojs23 [pie-1p::GFP::C34B2.10]</i>
FM302	CB4108	<i>fog-2(q71) V</i>
FM498	BN580	<i>baf-1(bq12[gfp::baf-1]) III</i>
FM500		<i>baf-1(bq12[gfp::baf-1]) III; itls37 [pie-1p::mCh::H2B::pie-1 3'UTR + unc-119(+)] IV</i>
FM539	JU2083	<i>Caenorhabditis macrosperma</i> wild isolate
FM602	JJ2586	<i>cox-4(zu476[cox-4::eGFP::3xFLAG]) I</i>
FM638		<i>ojs23 [pie-1p::GFP::C34B2.10]; wjls76[Cn_ unc-119(+); pie-1p::mKate2::tba-2]</i>
FM647	BCN9071	<i>vit-2(crg9070[vit-2::gfp]) X</i>
FM653	KWN724	<i>sdhc-1(jbm1 [sdhc-1::mCherry]) III; him-5(e1490) V</i>
FM727		<i>egxSi126 [mex-5p::hsp-3(aa1-19)::halotag::HDEL::pie-1 3'UTR + unc-119(+)] I; vit-2(crg9070[vit-2::gfp]) X</i>
FM862		<i>atx-2(syb5389; ATX-2::AID::GFP) III; ieSi38 [sun-1p::TIR1::mRuby::sun-1 3'UTR + Cbr-unc-119(+)] IV; wjls76[Cn_ unc-119(+); pie-1p::mKate2::tba-2]</i>
FM932		<i>duSi29(pFM1994[TMCO1::GFP(GLO)::SSPB(nanoGLO)]III} ; [pie-1p-mCh::PH(PLC1delta1) + unc-119(+)]V</i>
FM956		<i>ojs23; wjls76[Cn_ unc-119(+); pie-1p::mKate2::tba-2]; atx-2(syb5389; atx-2::AID::GFP); ieSi38 [sun-1p::TIR1::mRuby::sun-1 3'UTR + Cbr-unc-119(+)] IV</i>

Figure S1.

Distance of the sperm contents from the cortex of control embryos at metaphase I vs anaphase I.



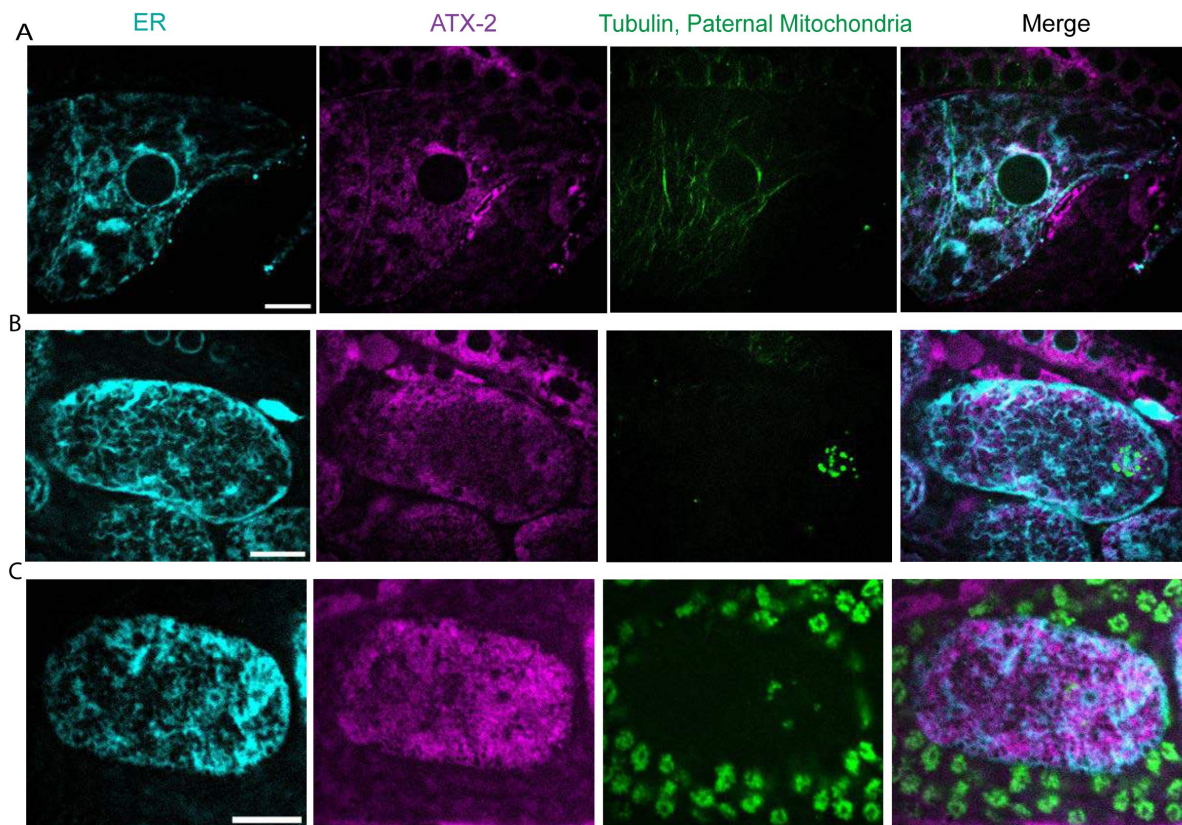


Figure S2

Localization of ATX-2 in -1 oocyte and +1 meiotic embryo.

Deconvolved single plane images from z-stacks acquired on a spinning-disk confocal. A. -1 oocyte (n=10). B. Metaphase meiotic embryo (n=10). C. Anaphase meiotic embryo (n=4). ER labeled with HALO-tag with the signal peptide and ER retention signal from HSP-3. Endogenous ATX-2::AID::GFP. mKate::Tubulin. Paternal mitochondria labeled with SDHC-1::mCherry. All Bars= 10 μ m.

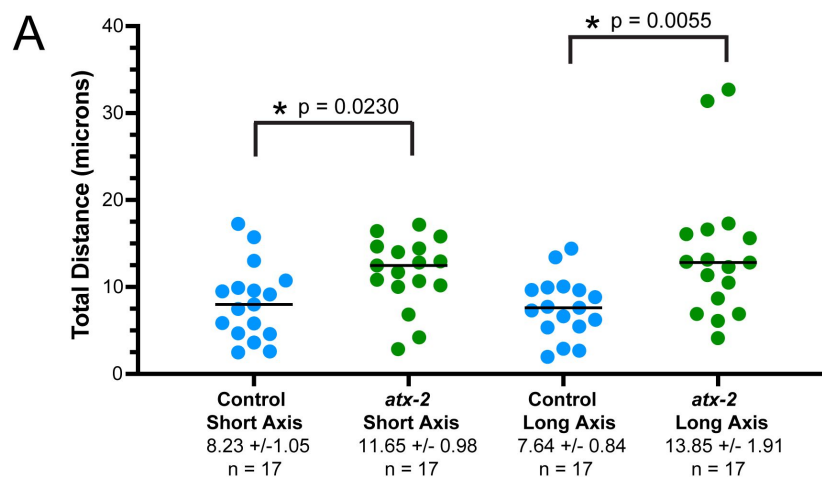


Figure S3.

Cytoplasmic streaming after ATX-2 depletion.

Maximum displacement of the sperm contents during any meiotic cell-cycle phase in ATX-2::AID::GFP embryos with or without 1 hr auxin.

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Editors

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Joint Public Review:

Summary:

This paper by Beath et. al. identifies a potential regulatory role for proteins involved in cytoplasmic streaming and maintaining the grouping of paternal organelles: holding sperm contents in the fertilized embryos away from the oocyte meiotic spindle so that they don't get ejected into the polar body during meiotic chromosome segregation. The authors show that by time-lapse video, paternal mitochondria (used as a readout for sperm and its genome) is excluded from yolk granules and maternal mitochondria, even when moving long distances by cytoplasmic streaming. To understand how this exclusion is accomplished, they first show that it is independent of both internal packing and the engulfment of the paternal chromosomes by the maternal endoplasmic reticulum creating an impermeable barrier. They then test whether the control of cytoplasmic streaming affects this exclusion by knocking down two microtubule motors, Katanin and kinesin I. They find that the ER ring, which is used as a proxy for paternal chromosomes, undergoes extensive displacement with these treatments during anaphase I and interacts with the meiotic spindle, supporting their hypothesis that the exclusion of paternal chromosomes is regulated by cytoplasmic streaming. Next, they test whether a regulator of maternal ER organization, ATX-2, disrupts sperm organization so that they can combine the double depletion of ATX-2 and KLP-7, presumably because *klp-7* RNAi (unlike *mei-1* RNAi) does not affect polar body extrusion and they can report on what happens to paternal chromosomes. They find that the knockdown of both ATX-2 and KLP-7 produces a higher incidence of what appears to be the capture of paternal chromosomes by the meiotic spindle (5/24 vs 1/25). However, this capture event appears to halt the cell cycle, preventing the authors from directly observing whether this would result in the paternal chromosomes being ejected into the polar body.

The authors addressed the vast majority of the Reviewer's comments including the addition of new figures, re-wording of data interpretation and discussion points to better reflect the claims of the paper. There remain a few outstanding points which were not addressed.

In many cases the number of embryos analyzed or events captured remains low and the authors conclude that these sample sizes prevented statistical significance. It's not clear if more embryos were analyzed or if more capture would lead to statistical significance. Language capturing this caveat should also be included in the manuscript. A specific example of this is given below:

In the double knockdown of ATX-2 and KLP-7, there was no significant difference between single and double knockdowns and the ER ring displacement was not analyzed in this double mutant. Further, there was no difference in the frequency of sperm capture between single and double ATX-2 and KLP-7 due to low sample size, the strength of the conclusion of this manuscript would be greatly improved if both of these results were further explored.

<https://doi.org/10.7554/eLife.97812.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

This paper by Beath et. al. identifies a potential regulatory role for proteins involved in cytoplasmic streaming and maintaining the grouping of paternal organelles: holding sperm contents in the fertilized embryos away from the oocyte meiotic spindle so that they don't get ejected into the polar body during meiotic chromosome segregation. The authors show that by time-lapse video, paternal mitochondria (used as a readout for sperm and its genome) is excluded from yolk granules and maternal mitochondria, even when moving long distances by cytoplasmic streaming. To understand how this exclusion is accomplished, they first show that it is independent of both internal packing and the engulfment of the paternal chromosomes by maternal endoplasmic reticulum creating an impermeable barrier. They then test whether the control of cytoplasmic streaming affects this exclusion by knocking down two microtubule motors, Katanin and kinesin I. They find that the ER ring, which is used as a proxy for paternal chromosomes, undergoes extensive displacement with these treatments during anaphase I and interacts with the meiotic spindle, supporting their hypothesis that the exclusion of paternal chromosomes is regulated by cytoplasmic streaming. Next, they test whether a regulator of maternal ER organization, ATX-2, disrupts sperm organization so that they can combine the double depletion of ATX-2 and KLP-7, presumably because klp-7 RNAi (unlike mei-1 RNAi) does not affect polar body extrusion and they can report on what happens to paternal chromosomes. They find that the knockdown of both ATX-2 and KLP-7 produces a higher incidence of what appears to be the capture of paternal chromosomes by the meiotic spindle (5/24 vs 1/25). However, this capture event appears to halt the cell cycle, preventing the authors from directly observing whether this would result in the paternal chromosomes being ejected into the polar body.

Strengths:

This is a useful, descriptive paper that highlights a potential challenge for embryos during fertilization: when fertilization results in the resumption of meiotic divisions, how are the paternal and maternal genomes kept apart so that the maternal genome can undergo chromosome segregation and polar body extrusion without endangering the paternal genome? In general, the experiments are well-executed and analyzed. In particular, the authors' use of multiple ways to knock down ATX-2 shows rigor.

Weaknesses:

The paper makes a case that this regulation may be important but the authors should do some additional work to make this case more convincing and accessible for those outside the field. In particular, some of the figures could include greater detail to support their conclusions, they could explain the rationale for some experiments better and they could perform some additional control experiments with their double depletion experiments to better support their interpretations. Also, the authors' inability to assess the functional biological consequences of the capture of the sperm genome by the oocyte spindle should be discussed, particularly in light of the cell cycle arrest that they observe.

These general comments are addressed in the more specific critiques below.

Reviewer #2 (Public Review):

Summary

*In this manuscript, Beath et al. use primarily *C. elegans* zygotes to test the overarching hypothesis that cytoplasmic mechanisms exist to prevent interaction between paternal chromosomes and the meiotic spindle, which are present in a shared zygotic cytoplasm after fertilization. Previous work, much of which by this group, had characterized cytoplasmic streaming in the zygote and the behavior of paternal components shortly after fertilization, primarily the clustering of paternal mitochondria and membranous organelles around the paternal chromosomes. This work set out to identify the molecular mechanisms responsible for that clustering and test the specific hypothesis that the "paternal cloud" helps prevent the association of paternal chromosomes with the meiotic spindle.*

Strengths

This work is a collection of technical achievements. The data are primarily 3- and 4-channel time-lapse images of zygotes shortly after fertilization, which were performed inside intact animals. There are many instances in which the experiments show extreme technical skill, such as tracking the paternal chromosomes over large displacements throughout the volume of the embryo. The authors employ a wide variety of fluorescent reporters to provide a remarkably clear picture of what is going on in the zygote. These reagents and the novel characterization of these stages that they provide will be widely beneficial to the community.

The data provide direct visualization of what had previously been a mostly hypothetical structure, the "paternal cloud," using simultaneous labeling of paternal DNA and mitochondria in combination with a variety of maternal proteins including maternal mitochondria, yolk granules, tubulin, and plasma membrane. Together, these images provided convincing evidence of the existence of this specified cytoplasmic domain. They go on to show that the knockdown of the ataxin-2 homolog ALX-2, a protein previously shown to affect ER dynamics, disrupted the paternal cloud, identifying a role for ER organization in this structure.

The authors then used the system to test the functional consequences of perturbing the cytoplasmic organization. Consistent with the paternal cloud being a stable structure, it stayed intact during large movements the authors generated using previously published knockdowns (of mei-1/katanin and kinesin-13/kpl-7) that increased cytoplasmic streaming. They used this data to document instances in which the paternal chromosomes were likely to have been attached to the spindle. They concluded with direct evidence of spindle fibers connecting to the paternal chromatin upon knockdown of ATX-2 in combination with increased cytoplasmic streaming, providing strong, direct support for their overarching hypothesis.

Weaknesses

While the data is convincing, the narrative of the paper could be streamlined to highlight the novelty of the experiments and better articulate the aims. For example, the cloud of paternal mitochondria and membranous organelles was previously shown, but Figures 1-2 largely reiterate that observation. The innovation seems to be that the combination of ER, yolk, and maternal mitochondrial markers makes the existence of a specified domain more concrete. There are also some instances where more description is needed to make the conclusions from the images clear.

These general comments are addressed in the more specific critiques below.

The manuscript intersperses what read like basic characterizations of fluorescent markers that, as written, can distract from the main story. The authors characterized the dynamics of ER organization throughout the substages of meiosis and the permeability of the envelope of ER that surrounds the paternal chromatin, but it could be more clearly established how the ability to visualize these structures allowed them to address their aims.

We have added the following after the initial description of ER morphology changes: (ER morphology was used to determine cell-cycle stages during live imaging reported below in Fig. 6.)

More background on what was previously known about ER organization in M-phase and the role of ataxin proteins specifically may help provide more continuity.

We have added references to transitions to ER sheets during mitotic M-phase in HeLa cells and *Xenopus* extracts.

Reviewer #3 (Public Review):

Summary:

This study by Beath et al. investigated the mechanisms by which sperm DNA is excluded from the meiotic spindle after fertilization. Time-lapse imaging revealed that sperm DNA is surrounded by paternal mitochondria and maternal ER that is permeable to proteins. By increasing cytoplasmic streaming using kinesin-13 or katanin RNAi, the authors demonstrated that limiting cytoplasmic streaming in the embryo is an important step that prevents the capture of sperm DNA by the oocyte meiotic spindle. Further experiments showed that the Ataxin-2 protein is required to hold paternal mitochondria together and close to the sperm DNA. Finally, double depletion of kinesin-13 and Ataxin-2 suggested an increased risk of meiotic spindle capture of sperm DNA.

Overall, this is an interesting finding that could provide a new understanding of how meiotic spindle capture of sperm DNA and its accidental expulsion into the polar body is prevented. However, some conceptual gaps need to be addressed and further experiments and improved data analyses would strengthen the paper.

- It would be helpful if the authors could discuss in good detail how they think maternal ER surrounds the sperm DNA

We have added 2 references to papers about nuclear envelope re-assembly from Shirin Bahmanyar's lab and suggest the ER envelope is a halted intermediate in nuclear envelope reassembly.

and why is it not disrupted following Ataxin disruption.

We have been attempting to disrupt ER structures in the meiotic embryo for the last 5 years by depleting profilin, BiP, atlastin, ATX-2 and by optogenetically packing ER into a ball in the middle of the oocyte. None of these treatments prevent envelopment of the sperm DNA by maternal ER. None of these treatments remove ER from the spindle envelope and none remove ER from the plasma membrane. These treatments mostly result in "large aggregates" of ER that we have not examined by EM. Wild speculation: any disruption of the ER strong enough to prevent ER envelopment around chromatin would be sterile because the M to S transition in the mitotic zone of the germline would be blocked. Rapid depletion of ATX-2 to the extent shown by rigorous data in this manuscript does not prevent ER envelopment

around chromatin. We chose not to speculate about the reasons for this because we do not know why.

- Since important phenotypes revealed in RNAi experiments (e.g. kinesin-13 and ataxin-2 double depletion) are not very robust, the authors should consider toning down their conclusions and revising some of their section headings. I appreciate that they are upfront about some limitations, but they do nonetheless make strong concluding sentences.

We have changed the discussion of the *klp-7 atx-2* double depletion to: “The capture of the sperm DNA by the meiotic spindle in ATX-2 KLP-7 double depleted embryos suggests that the integrity of the exclusion zone around the sperm DNA might insulate the sperm DNA from spindle microtubules. However, a much larger number of *klp-7(RNAi)* singly depleted and *atx-2(degron)* singly depleted time-lapse sequences are needed to rigorously support this idea.

- The discussion section could be improved further to present the authors' findings in the larger context of current knowledge in the field.

We have expanded the discussion as suggested.

- The authors previously demonstrated that F-actin prevents meiotic spindle capture of sperm DNA in this system. However, the current manuscript does not discuss how the katanin, kinesin-13 and Ataxin-2 mechanisms could work together with previously established functions of F-actin in this process.

We have added *pfn-1(RNAi)* to the discussion section.

- How can the authors exclude off-target effects in their RNAi depletion experiments? Can kinesin-13, katanin, and Ataxin phenotypes be rescued for instance?

For ataxin-2 phenotypes, two completely independent controls for off target effects are shown. GFP(RNAi) on a strain with and endogenous ATX-2::GFP tag vs GFP(RNAi) on a strain with no tag on the ATX-2. ATX-2::AID with or without auxin. For kinesin-13 and katanin, we did not do a rigorous control for off-target effects of RNAi. However, the effects of these depletions on cytoplasmic microtubules have been previously reported by others

- How are the authors able to determine if the paternal genome was actually captured by the spindle? Does lack of movement definitively suggest capture without using a spindle marker?

mKate::tubulin labels the spindle in each capture event. This can be seen in Video S3. for *mei-1(RNAi)* and Figure 9 for *atx-2 klp-7* double depletions.

(1) Major issues:

The images provided are not convincing that mitochondria are entirely excluded from the regions with yolk granules from the images provided. Please provide insets of magnified images of the paternal mitochondria in Figure 1E to more clearly show the exclusion even when paternal mitochondria are streaming. Providing grayscale images, individual z-sections and/or some quantification of this data might also be more convincing to this reviewer.

We have modified Fig. 1 by adding single wavelength magnified insets to more clearly show that paternal mitochondria are in a “black hole” in the maternal yolk granules during cytoplasmic streaming.

Figure 2 - This figure can be retitled to highlight that the paternal organelle cloud is impermeable to mitochondria and conserved.

The legend has been re-titled as suggested.

Figure 3B, An image of the DNA within the ring of maternal ER especially since the maternal ER ring is used as a proxy for the paternal chromosomes in later figures would strengthen the authors' claims.

We have added a panel showing DAPI-stained DNA in the center of the ER ring and paternal mitochondria cloud.

Why is the faster time scale imaging significant? I think this could be more clearly set up in the paper. Perhaps rapid imaging of maternal mito-labeled kca-1(RNAi) embryos would better show the difference in time scale, with the expectation that the paternal cloud forms and persists while the ER invades.

We are not sure what the reviewer means. 5 sec time intervals were used throughout the paper. We are also not sure how kca-1(RNAi) would help. Movement of the entire oocyte into and out of the spermatheca is what limits the ability to keep a fusing sperm in focus. kca-1(RNAi) would prevent cytoplasmic streaming but not ovulation movements.

Figure 4 - The question about the permeability of the ER envelope seems to come out of nowhere as written. It isn't clear how it contributes to the larger story about preventing sperm incorporation in the spindle.

This section of the results is introduced with: “If the maternal ER envelope around sperm DNA was sealed and impermeable during meiosis, this could both prevent the sperm DNA from inducing ectopic spindle assembly and prevent the sperm DNA from interacting with meiotic spindle microtubules.”

The data in Figure 4 would probably not be expected to be in this paper based on the paper title. Maybe the title needs something about ER dynamics? "eg. ATX-2 but not an ER envelope" isolates the paternal chromatin?

In Figure 5, it seems that RNAi of klp-7 and Mei-1 had slightly different effects on short-axis displacement of the ER envelope (klp-7 affecting it more dramatically than mei-1) and slightly different effects on interaction with the meiotic spindle (capture vs streaming past the spindle). The authors mention in their discussion that the difference in the interaction with the meiotic spindle might reflect the effects that loss of Mei-1 may have on the spindle but could it also be a consequence of the differences in cytoplasmic streaming observed?

With our current data, the only statistically significant difference between cytoplasmic streaming of the sperm contents in mei-1(RNAi) vs klp-7(RNAi) is that excessive streaming persists longer into metaphase II in klp-7(RNAi). We have added a sentence describing this difference to the results. If differences in streaming were the cause of different capture frequencies, then klp-7(RNAi) would cause more capture events than mei-1(RNAi) but the opposite was observed. We have avoided too much discussion here because the frequency of capture events is too low to demonstrate statistically significant differences between mei-

1(RNAi), klp-7(RNAi), and atx-2(degron) + klp-7(RNAi) without a very large increase in the number of time-lapse sequences.

Also, the authors should find a way to represent this interaction with the meiotic spindle in a quantitative or table form to allow the reader to observe some of the patterns they report more easily.

We have added a table to Fig. 9 that summarizes capture data.

Finally, can the authors report when they observe the closest association with the meiotic spindle: Does it correlate with the period of greatest displacement (AI) or are they unlinked?

The low frequency of capture events makes it difficult to test this rigorously.

Figure 6- 'Endogenously tagged ATX-2 was observed throughout oocytes and meiotic embryos without partial co-localization with ER.' How can the authors exclude co-localization with ER?

We have changed the wording to: “Endogenously tagged ATX-2 was observed throughout oocytes and meiotic embryos (Fig. 6A; Fig. S2). ATX-2 did not uniquely co-localize with ER (Fig. S2).”

The rationale for why the authors think that the integrity of sperm organelles is important to keep the genomes apart is not clear to this reviewer and needs to be explained better. Moving the discussion of the displacement experiments in Figure S3 from the end of the results section to the ATX-2 knockdown section would help accomplish this.

We have added the sentence: “The frequency of sperm capture by the meiotic spindle (Fig. 9D) was significantly higher than wild-type controls in *klp-7(RNAi) atx-2(AID)* double depleted embryos ($p=0.011$ Fisher’s exact test). Although the number of single mutant embryos analyzed was too low to demonstrate a significant difference between single and double mutant embryos, these results qualitatively support the hypothesis that limiting cytoplasmic streaming and maintaining the integrity of the ball of paternal mitochondria are both important for preventing capture events between the meiotic spindle and sperm DNA.”

It looks like, in the double knockdown of ATX-2 and KLP-7, the spread of paternal mitochondria is less affected than when only ATX-2 is depleted. What effect does this result have on the observation that the incidence of sperm capture appears to increase in the double depletion? What does displacement of the ER ring look like in the double depletion? Is it additive, consistent with their interpretation that both limiting cytoplasmic streaming and maintaining the integrity of the ball of paternal mitochondria is required to keep the genomes separate?

We cannot show a significant difference between single a double knockdowns without increasing n by alot. We did not analyze ER ring displacement in the double mutant.

Is the increased incidence of capture in the double-depleted embryos significant?

We have added the sentence: “The frequency of sperm capture by the meiotic spindle (Fig. 9D) was significantly higher than wild-type controls in *klp-7(RNAi) atx-2(AID)* double depleted embryos ($p=0.011$ Fisher’s exact test). Although the number of single mutant embryos analyzed was too low to demonstrate a significant difference between single and double

mutant embryos, these results qualitatively support the hypothesis that limiting cytoplasmic streaming and maintaining the integrity of the ball of paternal mitochondria are both important for preventing capture events between the meiotic spindle and sperm DNA.”

What do the authors make of the cell cycle arrest observed when paternal chromosomes are captured? Is there an argument to be made that this arrest supports the idea that preventing this capture is actively regulated and therefore functionally important?

We chose not to discuss the mechanism of this arrest because considerably more work would be required to prove that it is not caused by a combination of imaging conditions and genotype. The low frequency of these capture + arrest events would make it very difficult to show that the arrest does not occur after depleting a checkpoint protein.

(2) Minor concerns:

Top of page 4: "streaming because depletion tubulin stops cytoplasmic streaming (7)" should be "streaming because depletion of tubulin stops cytoplasmic streaming (7)"

The "of" has been inserted.

Page 6: "This result indicated that the volume of paternal mitochondria excludes maternal mitochondria and yolk granules but not maternal ER." The authors have only shown this for maternal mitochondria, not yolk granules.

We have deleted the mention of yolk granules here.

Page 7: "These results suggest that all maternal membranes are initially excluded from the sperm at fusion." Should be "These results show that maternal ER are initially excluded from the sperm at fusion. Since maternal mitochondria and yolk granules are excluded later, this suggests that all maternal membranes are initially excluded from the sperm at fusion."

We have changed this sentence as suggested.

It's not clear why the authors show other types of movement that might be quantified when cytoplasmic streaming is affected in Figure 5A and only quantify long-axis and short-axis displacement.

We have deleted the other types of movement from the schematic. Although these parameters were quantified, we did not include this data in the results so it would be confusing for the reader to have them in the schematic.

Bottom of page 7: Mention that the GFP::BAF-1 was maternally provided.

We have added "Maternally provided.."

Missing an Arrow on Figure 1A 9:20.

We removed the text citation to an arrow in Fig. 1A because we moved most of the description of the ER ring to Fig. 3 to address other reviewer suggestions.

Supplemental videos should be labeled appropriately to indicate what structures are labeled. It is currently difficult to understand what is being shown.

(3) Issues with the Discussion section:

"The simplest explanation is that cytoplasm does not mix during the 45 min from GVBD to pronucleus formation due to the high viscosity of cytoplasm." - Citation page 12.

We have changed the sentence to: "The simplest hypothesis is that maternal and paternal cytoplasm might not mix during the 45 min from GVBD to pronucleus formation due to the high viscosity of cytoplasm."

"The higher frequency of capture of the sperm DNA by the meiotic spindle in ATX-2 KLP-7 double depleted embryos compared with either single depletion suggests that the integrity of the exclusion zone around the sperm DNA may insulate the sperm DNA from spindle microtubule" - Pages 12-13 reference the figures.

This sentence has been rewritten in response to other comments but the new sentence now references revised Fig. 9.

"ATX-2 is required to maintain the integrity of the ball of paternal mitochondria around the sperm DNA, but the mechanism is unknown." - Page 13 reference figure.

A reference to Figs 7 and 8 has been inserted.

"In control embryos, the sperm contents rarely came near the meiotic spindle in agreement with a previous study that found that male and female pronuclei rarely form next to each other (6). Streaming of the sperm contents was most commonly restricted to a jostling motion with little net displacement, circular streaming in the short axis of the embryo, or long axis streaming in which the sperm turned away from the spindle before the halfway point of the embryo. Depletion of MEI-1 or KLP-7 resulted in longer excursions of the sperm contents in the long axis of the embryo toward the spindle but frequent capture of the sperm by the spindle was only observed in mei-1(RNAi)." - Page 13, the corresponding figures need to be referenced for these sentences.

We have inserted figure references.

"In capture events observed after double depletion of ATX-2 and KLP-7, a bundle of microtubules was discernible extending from the spindle into the ER envelope surrounding the sperm DNA. Such bundles were not observed in mei-1(RNAi) capture events, likely because of the previously reported low density of microtubules in mei-1(RNAi) spindles (36, 37)." - Pages 13-14 references figures here.

We have inserted figure references.

"The higher frequency of capture of the sperm DNA by the meiotic spindle in ATX-2 KLP-7 double depleted embryos compared with either single depletion suggests that the integrity of the exclusion zone around the sperm DNA may insulate the sperm DNA from spindle microtubules." - This should be toned down since this phenotype is not robust.

We have changed this to: "The capture of the sperm DNA by the meiotic spindle in ATX-2 KLP-7 double depleted embryos suggests that the integrity of the exclusion zone around the sperm DNA might insulate the sperm DNA from spindle microtubules. However, a much larger number of *kfp-7(RNAi)* singly depleted and *atx-2(degron)* singly depleted time-lapse sequences are needed to rigorously support this idea. "

ATX-2 depletion alters ER morphology but does not impact the maternal ER envelope - could the authors provide a potential explanation for this?

In the discussion, we cite papers showing that ATX-2 depletion affects many different cellular processes so the effect we see on paternal mitochondria might have nothing to do with the ER ring. We have been attempting to disrupt ER structures in the meiotic embryo for the last 5 years by depleting profilin, BiP, atlastin, ATX-2 and by optogenetically packing ER into a ball in the middle of the oocyte. None of these treatments prevent envelopment of the sperm DNA by maternal ER. None of these treatments remove ER from the spindle envelope and none remove ER from the plasma membrane. These treatments mostly result in “large aggregates” of ER that we have not examined by EM. Wild speculation: any disruption of the ER strong enough to prevent ER envelopment around chromatin would be sterile because the M to S transition in the mitotic zone of the germline would be blocked. Rapid depletion of ATX-2 to the extent shown by rigorous data in this manuscript does not prevent ER envelopment around chromatin. We chose not to speculate about the reasons for this because we do not know why.

It would be good to have representative images of what the altered spindle looks like in MEI-1-depleted oocytes.

The structure of MEI-1-depleted spindles has been described in the cited references.

"Depletion of MEI-1 or KLP-7 resulted in longer excursions of the sperm contents in the long axis of the embryo toward the spindle but frequent capture of the sperm by the spindle was only observed in mei-1(RNAi)" - It is intriguing that this does not happen in the double depletion experiments of kinesin-13 and ATX-2. The authors should perhaps discuss this.

This does happen in KLP-7 ATX-2 double depleted embryos as shown in Fig. 9.

(4) Missing citations:

"This analysis was restricted to embryos from anaphase I through anaphase II because our streaming data and that of Kimura 2020 indicate that the sperm contents have not moved significantly before anaphase I." - This needs an appropriate citation. Page 10.

We have inserted citations here.

"The simplest explanation is that cytoplasm does not mix during the 45 min from GVBD to pronucleus formation due to the high viscosity of cytoplasm." - Citation page 12. Not referencing figures in the discussion.

We have changed the sentence to: “The simplest hypothesis is that maternal and paternal cytoplasm might not mix during the 45 min from GVBD to pronucleus formation due to the high viscosity of cytoplasm.”

"The higher frequency of capture of the sperm DNA by the meiotic spindle in ATX-2 KLP-7 double depleted embryos compared with either single depletion suggests that the integrity of the exclusion zone around the sperm DNA may insulate the sperm DNA from spindle microtubule" - Pages 12-13 reference the figures.

A reference to the revised Fig. 9 has been inserted in the revised version of this sentence.

"ATX-2 is required to maintain the integrity of the ball of paternal mitochondria around the sperm DNA, but the mechanism is unknown."

References to Figs. 7 and 8 have been inserted.

Page 13 reference figure

"In control embryos, the sperm contents rarely came near the meiotic spindle in agreement with a previous study that found that male and female pronuclei rarely form next to each other (6). Streaming of the sperm contents was most commonly restricted to a jostling motion with little net displacement, circular streaming in the short axis of the embryo, or long axis streaming in which the sperm turned away from the spindle before the halfway point of the embryo. Depletion of MEI-1 or KLP-7 resulted in longer excursions of the sperm contents in the long axis of the embryo toward the spindle but frequent capture of the sperm by the spindle was only observed in mei-1(RNAi)." Page 13, the corresponding figures need to be referenced for these sentences.

We have inserted citations here.

"In capture events observed after double depletion of ATX-2 and KLP-7, a bundle of microtubules was discernible extending from the spindle into the ER envelope surrounding the sperm DNA. Such bundles were not observed in mei-1(RNAi) capture events, likely because of the previously reported low density of microtubules in mei-1(RNAi) spindles (36, 37)." Pages 13-14 references figures here.

We have inserted citations here.

(5) Referencing wrong figures in the text:

Figure 5 - In the figure legend there is a 5C but there is no 5C panel in the figure.

A C has been inserted in Fig. 5.

Figure 6A - "Dark holes were observed suggesting exclusion from the lumens of larger membranous organelles (Fig. 6A; Fig. S2)." Page 10.

6A has been changed to 6C.

Figure 6A is showing background autofluorescence in WT oocytes so I am not certain why it is cited here.

The Figure citation has been corrected to 6B, C.

Figure 8 - I could not find the supplemental data file with the individual mitochondria distance measurements.

We are including the Excel file with the revised submission.

The last sentence of the first paragraph should be re-worded to be more concise ". In C. elegans, the nucleus is positioned away from the site of future fertilization so that the meiosis I spindle assembles at the opposite end of the ellipsoid zygote from the site of fertilization (2-4). "

Every word of this sentence is important.

Last sentence second paragraph typo "These microtubules are thought to drive meiotic cytoplasmic streaming because depletion tubulin stops cytoplasmic streaming (7) and depletion of the microtubule-severing protein katanin by RNAi results in an increased mass of cortical microtubules and an increase in cytoplasmic streaming (8)." Pages 3-4.

“of” has been inserted.

(6) Typos in the introduction should be corrected:

Ataxin or kinesin-13 are not mentioned in the introduction but these are a big focus of the paper.

Gong et al 2024 written instead of number citation (page 5), no citation in References.

This has been corrected.

Supplemental videos should be labeled appropriately to indicate what structures are labeled. It is currently difficult to understand what is being shown.

<https://doi.org/10.7554/eLife.97812.2.sa0>